Personal Account

THE CHEMICAL RECORD

Enantioselective Total Syntheses of Cyathane Diterpenoids

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ABSTRACT: A Personal Account describing the enantioselective total syntheses of cyathane diterpenoids achieved in the Nakada group. A convergent approach to the cyathane scaffold, the [5-6-7] tricyclic carbon skeleton commonly found in cyathane diterpenoids, has been developed using the catalytic asymmetric intramolecular cyclopropanation (CAIMCP) and baker's yeast reduction. This approach has been successfully applied for the enantioselective total syntheses of (+)-allocyathin B_2, (–)-erinacine B, and (–)-erinacine E. The total synthesis of (–)-erinacine E has been achieved via the acyl group migratory intramolecular aldol reaction, which prevents the retro-aldol reaction and allows the construction of the strained structure. The highly efficient and stereoselective total syntheses of (–)-scabronines G, A, D, and (–)-episcabronine A have been achieved via the oxidative dearomatization/inverse electron demand Diels–Alder reaction cascade. Cascade reactions comprising three and five consecutive reactions were employed for the highly efficient total syntheses of (–)-scabronine A and (–)-episcabronine A, respectively. **DOI 10.1002/tcr.201402019**

Keywords: asymmetric synthesis, cascade reactions, cyathanes, natural products, total synthesis

Introduction

Since the isolation and structural elucidation of the first cyathins from bird nest fungi by Ayer and co-workers,[1a] numerous similar compounds have been isolated and characterized, including the cyathane skeleton, cyathins,[1] allocyathins,[1d–g,i,j,m,2] striatins,[1h,3] cyafrins,[1g,j,k,4] sarcodonins,[1m,n,2,5k] erinacines,[5] scabronines,[6] glaucopines,[7] cyrneines,[7c,8] and cyanthiwigins[9] (Figure 1). With some exceptions, all the cyathane diterpenoids possess an unusual [5-6-7] tricyclic carbon skeleton, including a *trans*-fused [6-7] ring system, as the cyathane scaffold. In addition, a common structural feature is the presence of two all-carbon quaternary stereogenic centers at its ring junctions.

The structural complexity and diversity of the cyathane diterpenoids arise from the different degree of oxidation around the five- and seven-membered rings, and in addition, at the C19 position of the sarcodonins and the C17 position of the scabronines. Striatins and erinacines possess unique structures categorized as xylose conjugates of cyathins and cyanthiwigins possess an *epi*-cyathane skeleton with the reversed C9 all-carbon quaternary stereogenic center.[9]

Some members of this large family of natural products display strong antibiotic activity, and erinacines and scabronines have been found to exhibit strong stimulating activity toward nerve growth factor (NGF) synthesis. NGF is a member of a family of small secreted proteins known as neurotrophins, which are important signaling molecules for the growth, maintenance, and survival of neural cells.[10]

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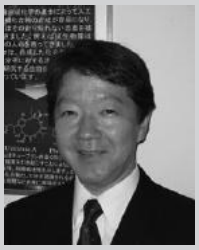
It has been suggested that administration of exogenous NGF could result in cytoprotection and stimulate the outgrowth of neuritic projections.[11] Therefore, direct administration of NGF to the brain has potential applicability in the treatment of neurodegenerative disorders,[12] but NGF does not cross the blood–brain barrier (BBB) and the native peptide is rapidly metabolized in vivo.[12f] Consequently, the use of small non-peptide stimulators of NGF synthesis has great potential as a possible treatment for neurodegenerative diseases such as Alzheimer's, Parkinson's, and Huntington's diseases.[12a,c,g]

It should be noted that erinacine E, one of the most complex members of this family of natural products, was recently shown to have not only potent NGF synthesis–stimulating activity but also κ-opioid receptor agonist activity.[5e] Furthermore, cyanthiwigins show toxicity against HBV, HIV-1, and human primary tumor cells.[9]

The structural complexity and diverse biological activity of cyathane diterpenoids have stimulated various research groups to investigate their synthesis,[13–16] leading to a number of total syntheses.[14,15]

In this Personal Account, our approaches to the enantioselective total syntheses of (+)-allocyathin B_2[14a] and (−)-erinacine B[14i] through a highly convergent and stereoselective construction of the [5-6-7] tricyclic core skeleton using novel chiral building blocks prepared by catalytic asymmetric intramolecular cyclopropanation (CAIMCP) and baker's yeast reduction, the total synthesis of (−)-erinacine E via a biomimetic cascade reaction,[14k] and the highly stereoselective total syntheses of (−)-scabronine G and A, (−)-episcabronine A,[14g] and (−)-scabronine D[14o] via cascade reactions will be described.



Masahisa Nakada received his B.S. (1982) and M.S. (1984) degrees from the University of Tokyo, and was appointed Assistant Professor there during his Ph.D. course in 1987. He received his Ph.D. degree in 1988 from the University of Tokyo under the supervision of Prof. Masaji Ohno and joined Prof. Shibasaki's group in 1991. After spending one year and four months from the beginning of FY1992 as a postdoctoral fellow with Prof. K. C. Nicolaou at the Scripps Research Institute, USA, he was promoted to Associate Professor at Waseda University in 1995. Since 2000, he has been a Professor in the Department of Chemistry and Biochemistry at Waseda University. He was awarded the Pharmaceutical Society Award for Young Scientists (1997) and SSOCJ Astellas Award for Organic Chemistry in Life Science (2008). His research interests include the total synthesis of complex bioactive natural products, asymmetric catalysis, new synthetic reactions, and chemical biology.

Total Synthesis of (+)-Allocyathin B_2[14a]

(±)-Allocyathin B_2 has succumbed to total synthesis by the Snider[15a,b] and Tori[15c] groups. However, no enantioselective total synthesis had been reported up until our synthesis.[14a] Scheme 1 shows our retrosynthetic analysis of (+)-allocyathin B_2.[14a] We envisaged that the keto ester **1** could be prepared from ketone **2** via a ring-expansion reaction. Installation of an ester group at the α position of the carbonyl group of ketone **2**, followed by iodomethylation, samarium diiodide mediated ring expansion, and subsequent introduction of the double bond would afford **1**. We envisaged that **2** could be prepared from diketone **3** via intramolecular aldol reaction, followed by dehydration, installation of the isopropyl group, and dehydration. Diketone **3** was disconnected to give the two chiral fragments **4** and **5**, which could be prepared from novel chiral building blocks previously reported by us, i.e., by CAIMCP[17] and baker's yeast reduction,[18a,b] respectively.

Preparation of the enantiomerically pure cyclopropane **6** via CAIMCP of the corresponding α-diazo-β-keto sulfone had been previously reported by us[17a] and was used to prepare **4** as outlined in Scheme 2.

Ring opening of cyclopropane **6** with benzenethiolate, followed by the selective removal of the mesityl sulfonyl group using lithium naphthalenide,[19] gave ketone **7** in 85% yield over two steps. Formation of the ethylene ketal derived from **7**, S-oxidation to the corresponding sulfoxide with *m*-chloroperbenzoic acid (*m*-CPBA) and subsequent Pummerer rearrangement gave **4** in 79% yield over three steps.

We previously reported the preparation of diol **8** using a baker's yeast mediated reduction of 2-benzyloxymethyl-2-methyl-1,3-cyclohexanedione and the stereoselective reduction using $Me_4NBH(OAc)_3$ (Scheme 3).[17a,b] Diol **8** was selected for the preparation of **5** because it possesses the all-carbon quaternary stereogenic center required for the enantioselective total synthesis of (+)-allocyathin B_2. Benzyl deprotection of diol **8** using hydrogenation and subsequent PMP-acetal formation afforded **9** as the sole product in 90% yield. TBS protection of the alcohol in **9** and subsequent regioselective reduction of the PMP-acetal using DIBAL-H afforded alcohol **10** in 97% yield over two steps. Compound **10** was converted to **5** in 93% yield using conventional iodination conditions.

With **4** and **5** in hand, coupling of these fragments was performed as outlined in Scheme 4. The organolithium compound generated upon treating **5** with *t*-BuLi in Et_2O was reacted with **4** to afford alcohol **11** in low yield. However, using a mixture of Et_2O/THF (10:1) as the reaction solvent greatly improved the yield of **11** to 79%. Alcohol **11** was converted to

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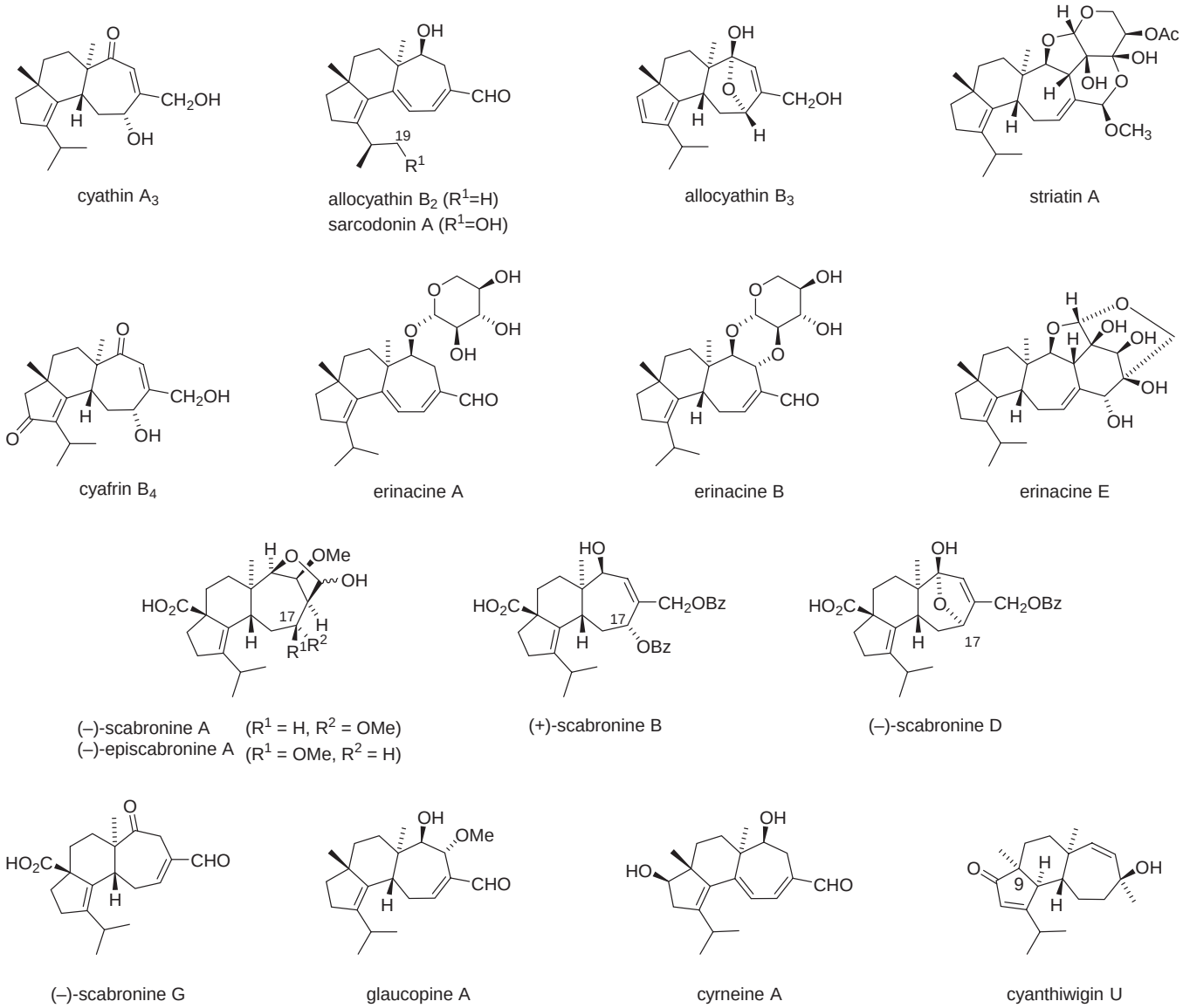
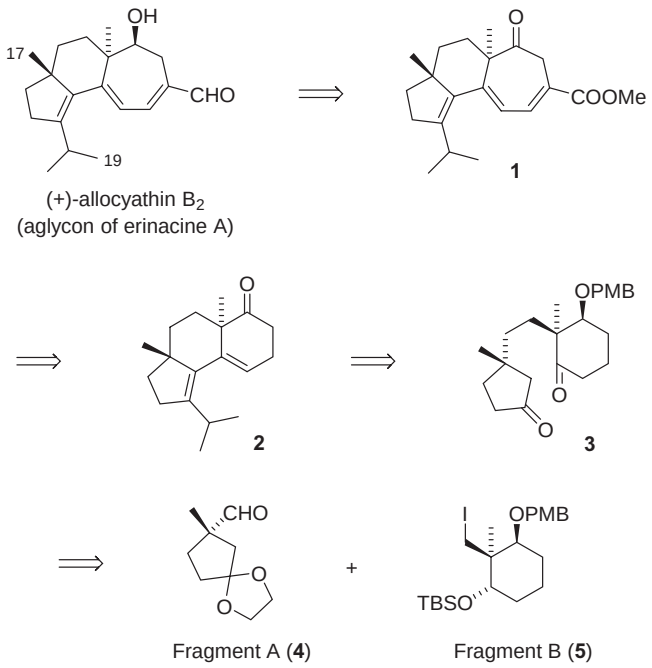


Fig. 1. Structures of selected cyathane diterpenoids.

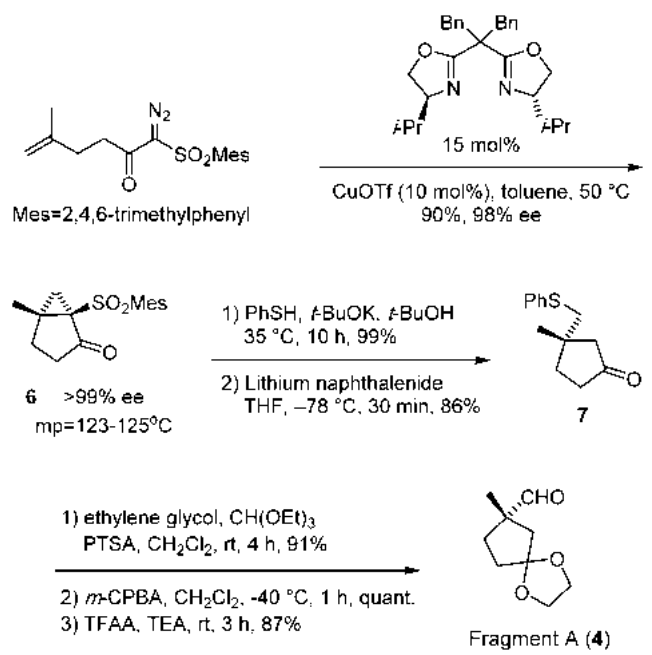
methyl xanthate **12** and reduced upon treatment with tin hydride[20] to give **13** in 81% yield over two steps. Both the protecting groups (TBS ether and ethylene ketal) in **13** were removed cleanly under the acidic conditions, followed by Dess–Martin oxidation to afford diketone **3** in 93% yield.

The intramolecular aldol reaction of **3** to form the six-membered ring in the cyathane skeleton was challenging because the reaction points in both the electrophile and the nucleophile are adjacent to the sterically hindered all-carbon quaternary stereogenic centers. Preliminary studies revealed that the aldol reaction using a base in a protic solvent gave an equilibrium mixture of diketone **3** and β-hydroxy ketone **14**, as the reaction was reversible under these conditions (retro-aldol reaction). For example, intramolecular aldol reaction of **3** using K_2CO_3 in ethanol at reflux temperature afforded a mixture of **3** (42%) and **14** (47%).

Accordingly, the aldol reaction was examined in an aprotic solvent. No reaction occurred with 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU) and 1,1,3,3-tetramethylguanidine; however, the use of potassium *t*-butoxide in benzene proved successful and the yield of **14** was improved to 94%. Molecular modeling suggested that this dramatic change could arise from the chelate that was formed between the hydroxyl and the keto groups in **14** with a potassium cation. The structure of **14** was elucidated by X-ray crystallographic analysis.[14a]

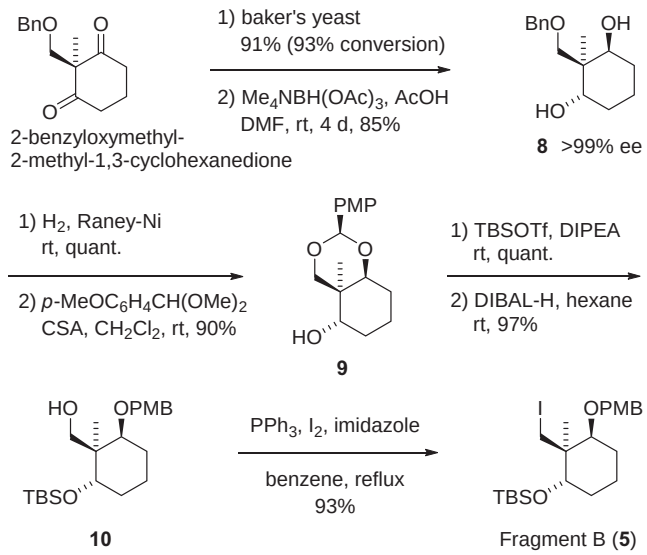


Scheme 1. Retrosynthetic analysis of (+)-allocyathin B_2.



Scheme 2. Preparation of fragment A (**4**).

Dehydration of β-hydroxy ketone **14** with *p*-toluenesulfonic acid (PTSA) (cat.) produced only the PMB deprotected enone **16** (45%). However, dehydration with thionyl chloride and pyridine as a base afforded a separable mixture of β,γ-unsaturated ketone **15** (71%) and its regioisomer **16**. Ketone **15** was reacted with isopropenyl lithium to afford alcohol **17**. Hydrogenation of the isopropenyl group took place with concomitant hydrogenolysis of the PMB group, and subsequent Dess–Martin oxidation of the resulting secondary alcohol afforded hydroxy ketone **18** in 88% yield over two steps. Dehydration of the hydroxy ketone **18** with thionyl chloride and pyridine as a base afforded a mixture of regioisomeric alkenes. However, acid-catalyzed alkene isomerization using PTSA in refluxing benzene gave the conjugated diene **2** in 85% yield.



Scheme 3. Preparation of fragment B (**5**).

Ring expansion of the cyclohexenone moiety in ketone **2** was accomplished using the method reported by Hasegawa[21] to generate the requisite γ-keto ester **20**. Compound **2** was converted to its corresponding β-keto ester using Mander's reagent,[22] and subsequent iodomethylation[23] afforded iodide **19** in 75% yield as a single isomer. Upon treating **19** with samarium diiodide, ring expansion proceeded to afford a 1.9:1 diastereoisomeric mixture of the desired γ-keto ester **20** in 91% yield.

To introduce the double bond into γ-keto ester **20**, conventional one-pot methods using $(PhSeO)_2O$[24] or *o*-iodoxybenzoic acid (IBX)[25] were attempted but proved to be unsuccessful. Consequently, we explored new reaction conditions for this transformation. After several attempts, the dienolate derived from **20** could be successfully converted into ketone **1**. The γ-keto ester **20** was treated with excess lithium diisopropylamide (LDA) in THF, followed by treatment with iodine, to afford ketone **1** in 71% yield. This one-pot procedure would involve isomerization from the initially formed α,β-unsaturated ketone to the more stable β,γ-unsaturated ketone **1** via the enolate. This transformation is clean and easy to execute, and therefore advantageous for preparative purposes.

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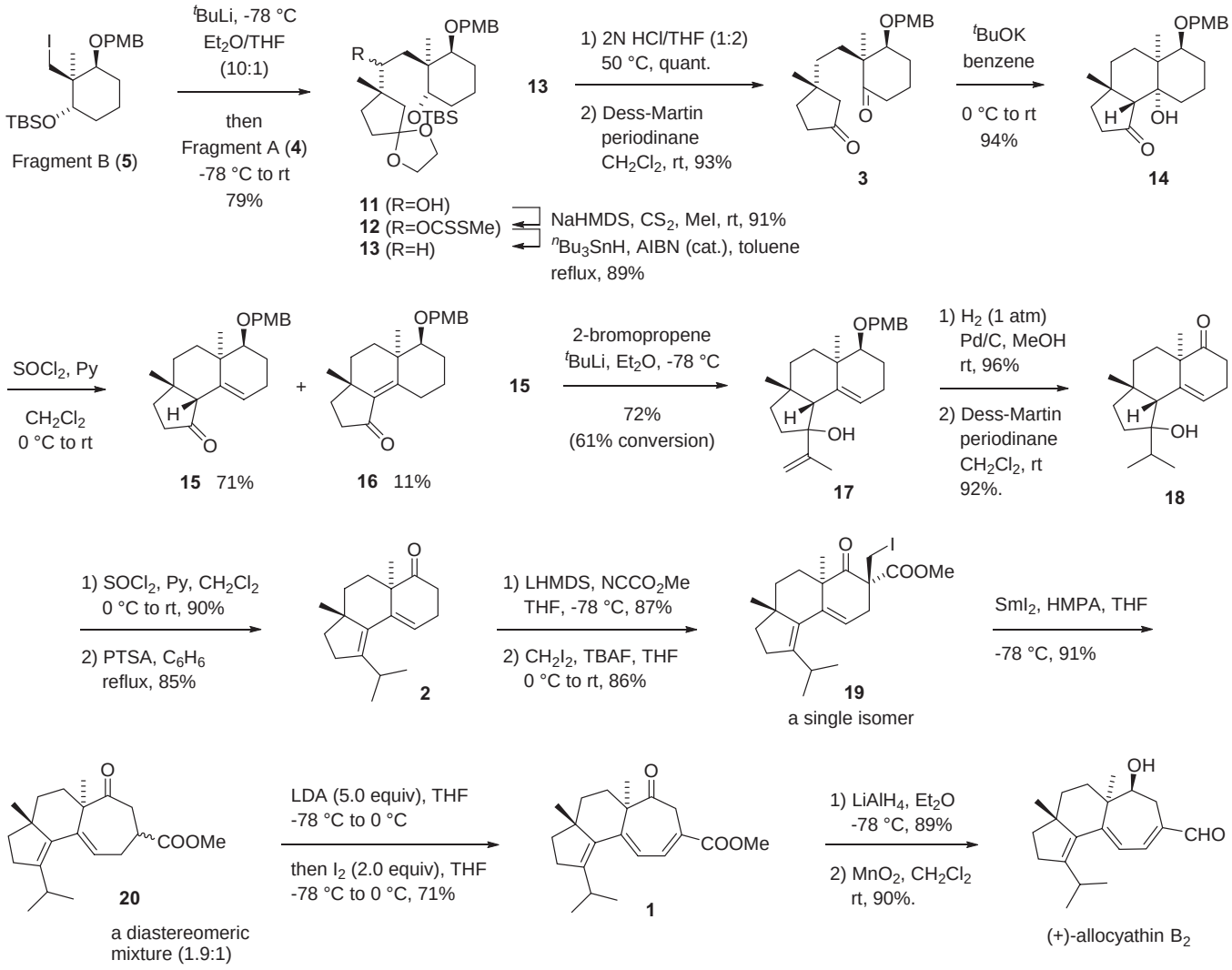


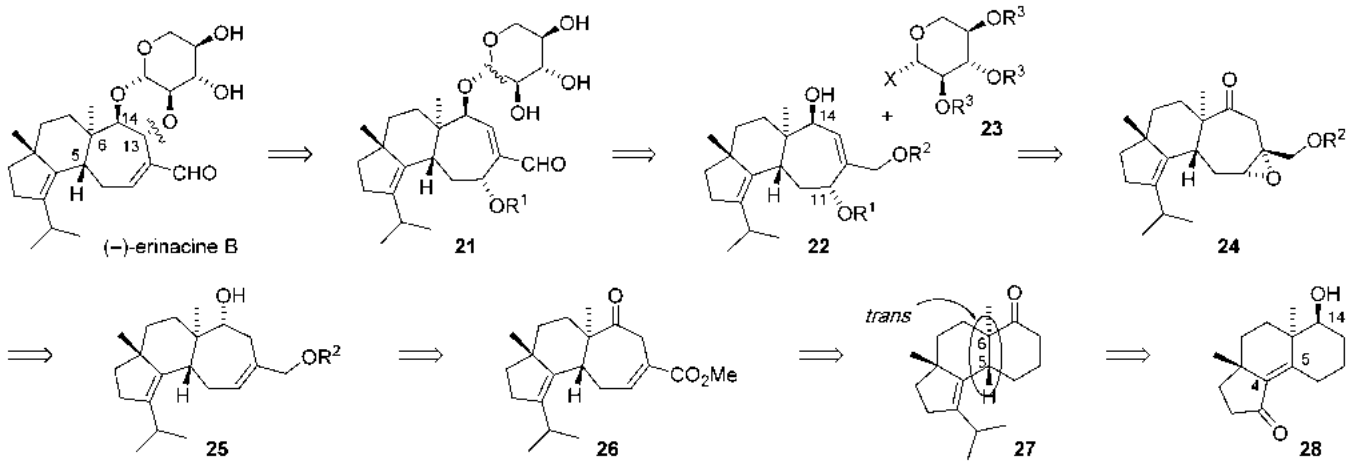
Scheme 4. Enantioselective total synthesis of (+)-allocyathin B_2.

Conversion of ketone **1** to allocyathin B_2 was accomplished using conventional transformations. A highly stereoselective reduction of the ketone and ester functionality in **1**, followed by selective oxidation of the resultant allylic alcohol with MnO_2, successfully completed the enantioselective total synthesis of (+)-allocyathin B_2. The synthetic (+)-allocyathin B_2 was identical in all respects to the natural product, as evidenced from the reported spectral data (^{1}H NMR,[15a,b] IR,[11] MS,[11] $[\alpha]_D$,[11] and ^{13}C NMR[15a,b]).

Our enantioselective total synthesis features a highly convergent synthesis of the [5-6-7] tricyclic core skeleton using the novel chiral building blocks prepared via asymmetric catalysis and an intramolecular aldol reaction, samarium diiodide mediated ring expansion, and a newly developed double-bond installation method as the key steps. With the novel synthetic approach to the [5-6-7] tricyclic core system of the cyathane diterpenoids established, we focused on the development of a new synthetic approach to the *trans*-fused [6-7] ring system present in the cyathane diterpenoids via an intermediate prepared in this synthesis.

Total Synthesis of (–)-Erinacine B[14i]

In 1994, the Kawagishi group reported the isolation and structural elucidation of (–)-erinacines A, B, and C.[5a] (–)-Erinacine B was the first xylose-conjugated terpenoid possessing a cyathane core to be identified, and it features an unusual [5-6-7] tricyclic carbon skeleton incorporating a *trans*-fused [6-7] ring system and 1,4-*anti* quaternary methyl groups at the



Scheme 5. Retrosynthetic analysis of (–)-erinacine B.

ring junctions. As described above, the erinacines and their congeners exhibit significant activity in stimulating NGF synthesis.

Our retrosynthetic analysis of (–)-erinacine B is shown in Scheme 5. (–)-Erinacine B was envisaged to be derived from xyloside **21** via an intramolecular 1,4-addition and β-elimination cascade reaction.[26] Xyloside **21** would be prepared via a glycosylation reaction of aglycon **22** with xylose derivative **23**. The preparation of **22** from epoxy ketone **24** was envisaged using a β-elimination reaction and stereoselective reduction of the resulting ketone. Epoxide **24** was expected to be prepared via stereoselective epoxidation of the homoallylic alcohol **25**, which itself could be prepared by the stereoselective reduction of ketone **26**.

We envisaged that the tricyclic compound **26** would be obtained from ketone **27** by a ring-expansion reaction and double-bond installation. The *trans*-C5–C6 stereodiad present in **27** could be generated by the diastereoselective reduction/protonation of enone **28** utilizing the β-oriented hydroxyl at C14 as the proton source.

Scheme 6 shows the preparation of compound **35** via enone **28**, which was prepared from the enantiopure β-hydroxy ketone **14**, whose synthesis was previously reported in the total synthesis of (+)-allocyathin B_2.[14a] Compound **14** was dehydrated using thionyl chloride and pyridine, followed by double bond isomerization using DBU and PMB deprotection with 2,3-dichloro-5,6-dicyano-*p*-benzoquinone (DDQ), to afford **28** in 84% yield over three steps.

We expected that the catalytic hydrogenation of **28** would proceed with high diastereoselectivity because of the directing effect of the β-oriented hydroxyl group at C14. However, hydrogenation of **28** was unsuccessful. Conjugate reduction of **28** with various reagents was also attempted, with little success; the desired product was not formed, or allylic alcohols corresponding to ketone reduction were observed.

We subsequently examined the electron-transfer reduction of **28** and found that the Birch reduction of **28** afforded a mixture of desired ketone **29** and its allylic alcohol. After rigorous optimization, reduction of **28** with samarium diiodide in the presence of HMPA proved successful and gave **29** as a single product in 94% yield.

The structure of ketone **29** was confirmed by the X-ray crystallographic analysis of the corresponding 3,5-dinitrobenzoate.[14i] It should be noted that this stereoselectivity was attributed to the diastereoselective protonation of the anion generated at C5 by the β-oriented hydroxyl group at C14, because reduction of the MOM ether of **28** under the same conditions gave the corresponding allylic alcohol exclusively.

Reaction of **29** with isopropenyl lithium gave low conversions. However, this problem was solved using the isopropenyl cerium reagent, and the resulting alcohol was subjected to hydrogenation, followed by Dess–Martin oxidation, to afford hydroxy ketone **30**. Dehydration of **30** using thionyl chloride and pyridine proceeded cleanly, and subsequent double bond isomerization under acidic conditions gave ketone **27** with a thermodynamically stable tetrasubstituted alkene.

The next task in the synthesis was to construct the [5-6-7] cyathane core from **27**, which required a one-carbon ring-expansion reaction. The ring-expansion method described by Hasegawa[21] was employed because it had been successfully applied in our enantioselective total synthesis of (+)-allocyathin B_2.[14a] Therefore, **27** was converted to the corresponding β-keto ester using Mander's reagent, and subsequent iodomethylation and treatment with samarium diiodide in the presence of HMPA gave the desired γ-keto ester **31** in 42% yield over three steps.

In our total synthesis of (+)-allocyathin B_2, introduction of the double bond to the seven-membered ring was achieved upon treatment with LDA and I_2, but in this case the yield was

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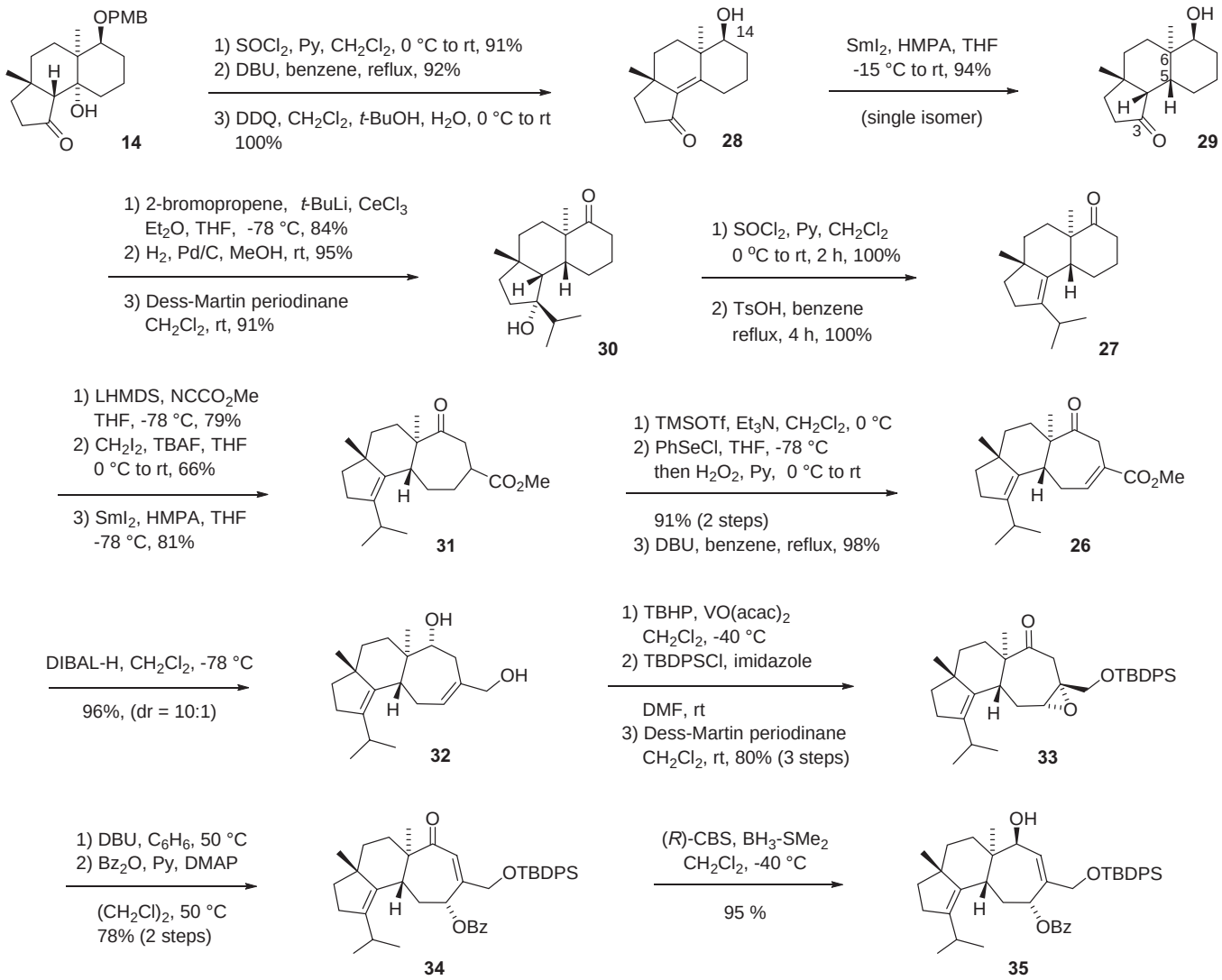


Scheme 6. Preparation of aglycon **35**.

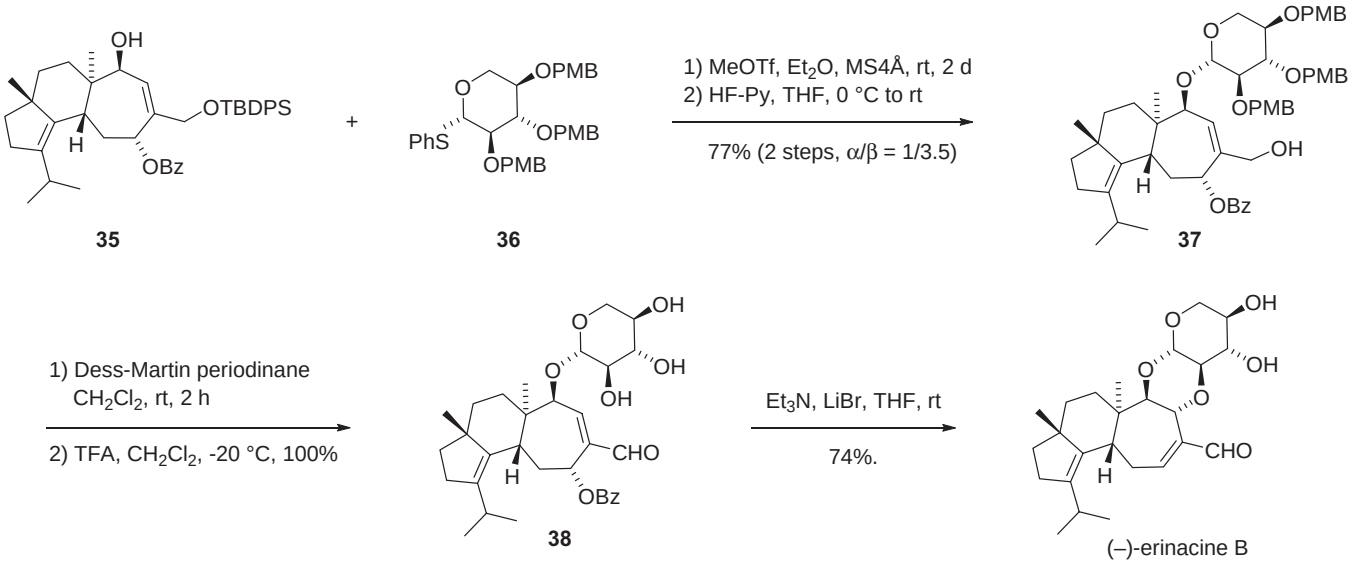
low. Fortunately, reaction of the TMS enol ether derived from the γ-keto ester **31** with PhSeCl afforded an α-phenylselenyl ketone, which was subjected to a selenoxide fragmentation reaction to successfully install the regioisomeric double bond that was easily isomerized using DBU to give the desired β-keto ester **26** in 89% yield over three steps.

DIBAL-H reduction of **26** exclusively provided the desired diol **32**, and stereoselective epoxidation using Sharpless conditions (TBHP/VO(acac)$_2$)[27] gave the desired epoxide as a single isomer. TBDPS protection of the primary alcohol and Dess–Martin oxidation of the remaining secondary alcohol afforded β,γ-epoxy ketone **33** in 80% yield over three steps from **32**.

Treatment of **33** with DBU prompted a clean β-elimination reaction to provide the γ-hydroxy α,β-unsaturated ketone, which was converted to benzoate **34** in 78% yield over two steps. The benzoyl group was chosen because we expected it to be stable during the subsequent transformations and that it would act as a good leaving group in the final intramolecular 1,4-addition/β-elimination cascade.

Diastereoselective reduction of the ketone in **34** using *substrate* control with a variety of achiral reagents was extensively investigated, but all of the reactions produced the undesired isomer as the major product. Consequently, we examined the use of a chiral *reagent*-controlled reduction reaction; the CBS-catalyzed reduction[28] of **34** proved successful and gave the desired alcohol **35** in 95% yield as a single isomer.

The glycosylation of **35** was expected to be difficult because the C14 hydroxyl group is highly sterically hindered



Scheme 7. Enantioselective total synthesis of (–)-erinacine B.

due to the all-carbon quaternary stereogenic center in the adjacent position. Most of the glycosylation reactions investigated using **35** with several xylose derivatives under various conditions did not give the desired glycosylated product.

However, we finally found that reacting **35** with thioglycoside **36** in the presence of MeOTf[29] gave the desired product, which underwent TBDPS deprotection upon treatment with HF·Py to give alcohol **37** in 77% yield as an inseparable mixture of anomeric isomers (Scheme 7).

Dess–Martin oxidation of alcohol **37** cleanly provided the corresponding aldehyde. However, subsequent removal of the PMB groups using DDQ was troublesome and afforded a complex mixture of products. After rigorous optimization, we found that treating the aldehyde with TFA[30] in CH_2Cl_2 at –20°C gave the key intermediate **38** in quantitative yield.

Finally, reaction of **38** with triethylamine and LiBr in THF using the procedure reported by Sassa[26] allowed the highly stereoselective oxa-Michael addition/β-elimination cascade reaction to occur and gave (–)-erinacine B in 74% yield as a single isomer. Synthetic (–)-erinacine B prepared using this route proved to be identical in all respects to the natural product, as seen from the spectral data.

Our total synthesis of (–)-erinacine B established a convergent synthesis of the xylose-conjugated [5-6-7] tricyclic cyathane core system bearing a *trans* [6-7] ring junction, using chiral building blocks prepared via the asymmetric catalysis route developed by us. Encouraged by this highly stereoselective total synthesis, we proceeded to study the enantioselective total synthesis of erinacine E, one of the most complex compounds in the family of xylose-conjugated cyathanes.

Total Synthesis of (–)-Erinacine E[14k]

In 1996 the Kawagishi group isolated (–)-erinacine E from the mycelia of *Hericium erinaceum*, a strong stimulator of NGF synthesis whose fruiting bodies have been used in Chinese medicine and as food (Figure 2).[5d] Two years later, a Pfizer research group isolated (–)-erinacine E from a different source, the fermentation broth of *H. ramosum* CL24240, and disclosed that (–)-erinacine E exhibited a potent, highly selective κ-opioid receptor agonist activity.[5e,31]

Single-crystal X-ray crystallographic analysis was used to elucidate the absolute structure of (–)-erinacine E. Its unique hexacyclic ring system comprises ten stereogenic centers on the [5-6-7] tricyclic cyathane skeleton, which is fused with a highly oxygenated tricyclic dioxatricycloundecane system. These unique structural features make (–)-erinacine E a strained and structurally different compound when compared with other cyathane diterpenoids and erinacines.[1–9]

Since the striatals, exemplified by striatal A and striatal D, and (–)-erinacine P (Figure 2) have been isolated,[1h;3,32] it was proposed that a C–C bond-forming reaction (***a***) between the C2′ and C13 positions in the keto aldehyde **39** could provide the striatals and a subsequent intramolecular aldol reaction (***b***) between the C4′ and C15 positions could provide the erinacines (Scheme 8). However, to the best of our knowledge, none of these transformations have been reported.

The complex structure, significant biological activity, and biogenesis of (–)-erinacine E make it a fascinating target. Since we had already achieved the total synthesis of (–)-erinacine B, we started to investigate the enantioselective total synthesis of (–)-erinacine E via alcohol **35**, which would be easily derived

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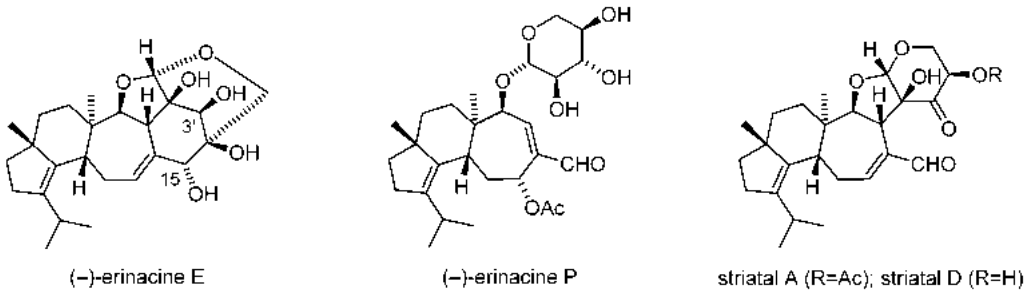
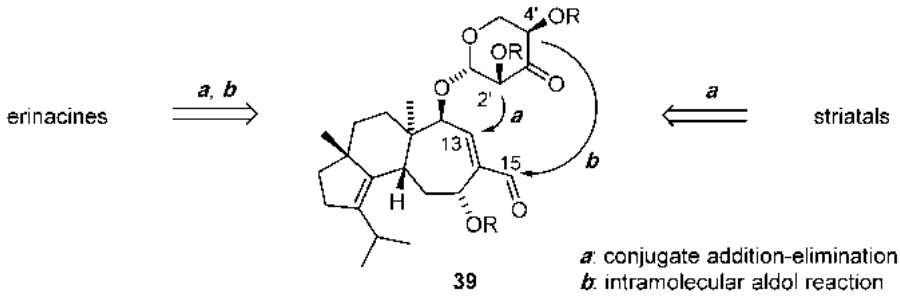


Fig. 2. Structures of (–)-erinacine E, striatals A and D, and (–)-erinacine P.



Scheme 8. Proposed biosynthesis of erinacines and striatals.

from an intermediate used in the total synthesis of (–)-erinacine B.

We first examined the glycosylation of alcohol **35** with thioglycoside **40**[14k] (Scheme 9). The glycosylation with MeOTf[14i,29] successfully afforded glycoside **41** in 84% overall yield with high selectivity ($\alpha/\beta = 1/14$) using the recycling technique. Specifically, the glycosylation for two days afforded **41** in 73% yield ($\alpha/\beta = 1/7$); however, the reaction quenched after one day afforded **41** in 55% yield with improved stereoselectivity ($\alpha/\beta = 1/14$), and recycling a recovered mixture of **35** and **40** afforded glycoside **41** in 29% yield ($\alpha/\beta = 1/14$). TBDPS and TES deprotection of **41** gave diol **42**, whose Swern oxidation afforded keto aldehyde **43** that underwent gradual conversion in situ to the PMB-protected striatal **44** due to base catalysis by Et_3N.

It was anticipated that the aldol reaction of **44** would be difficult to achieve because the product is highly strained, and thereby energetically unfavorable, and would be prone to undergo a retro-aldol reaction. Actually, extensive studies indicated that the aldol reaction of **44** under various reaction conditions resulted in either no reaction or decomposition.

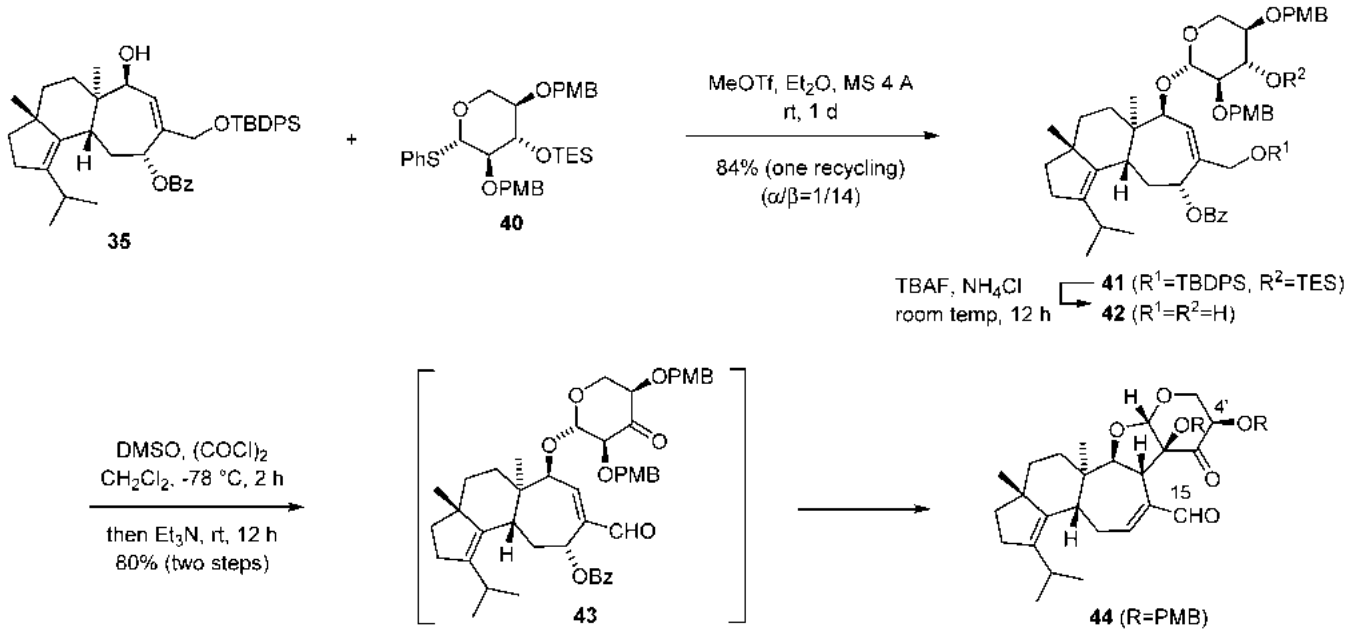
Considering the structure of striatal A (Figure 2), a proposed biosynthetic intermediate of (–)-erinacine E, we were inspired by the C4′ acetyl group and decided to examine the intramolecular aldol reaction of ketone **49** bearing the C4′ benzoate (see Scheme 11). A benzoyl group was used instead of the acetyl group because of its stability.

Deprotection of **44** caused decomposition under the reaction conditions used. Therefore, we prepared ketone **50** (see Scheme 11) from diol **42** (Scheme 10). PMB deprotection of **42** provided compound **45**, whose subsequent TES protection gave a separable 1:1 mixture of TES ethers **46** and **47** in 88% yield. Recycling of TES ether **47** to **45** was successfully achieved.

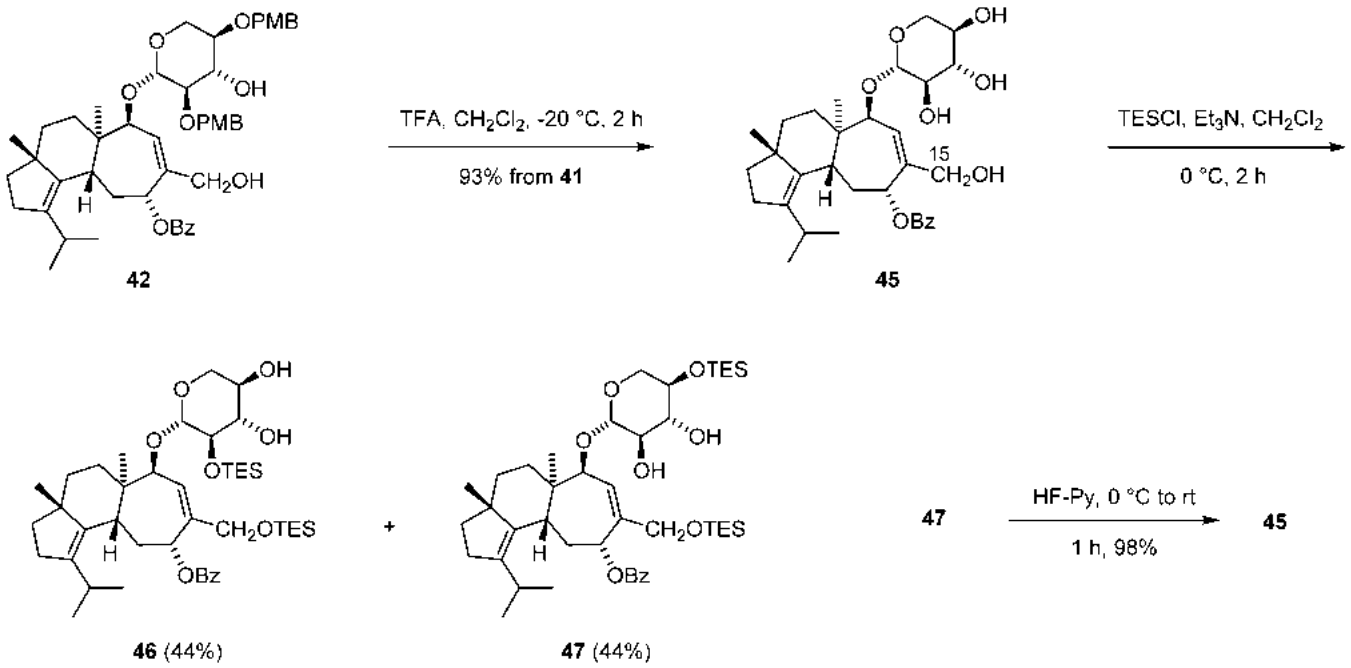
Selective benzoylation of the least hindered hydroxyl group in **46** (Scheme 11), deprotection of the C15 TES group, and Swern oxidation of the resulting diol gave the intermediate keto aldehyde **49**, which underwent cyclization/elimination in situ to give ketone **50** in 78% yield over three steps.

Initially, the intramolecular aldol reaction of **50** was attempted using bases containing a metal cation (such as *t*-BuOK, LiBr/Et_3N, and Cs_2CO_3), but no reaction or decomposition occurred. However, the reaction with DBU effectively afforded the product in 85% yield. Detailed studies of the product using NMR spectroscopy[14k] revealed its structure as ketone **53** and proved that the aldol reaction expectedly proceeded with the concomitant benzoyl group migration.

Reduction of **53** using $NaBH_4$ proceeded as anticipated, with addition from the less hindered face to give alcohol **54**, whose subsequent deprotection gave compound **55** in 96% yield over two steps. Mitsunobu reaction of **55** was unsuccessful, and all attempts at inverting the stereochemistry at C15 failed. Therefore, an oxidation–stereoselective reduction



Scheme 9. Enantioselective synthesis of PMB-protected striatal **44**.



Scheme 10. Preparation of diol **46** and recycling of **47** to **45**.

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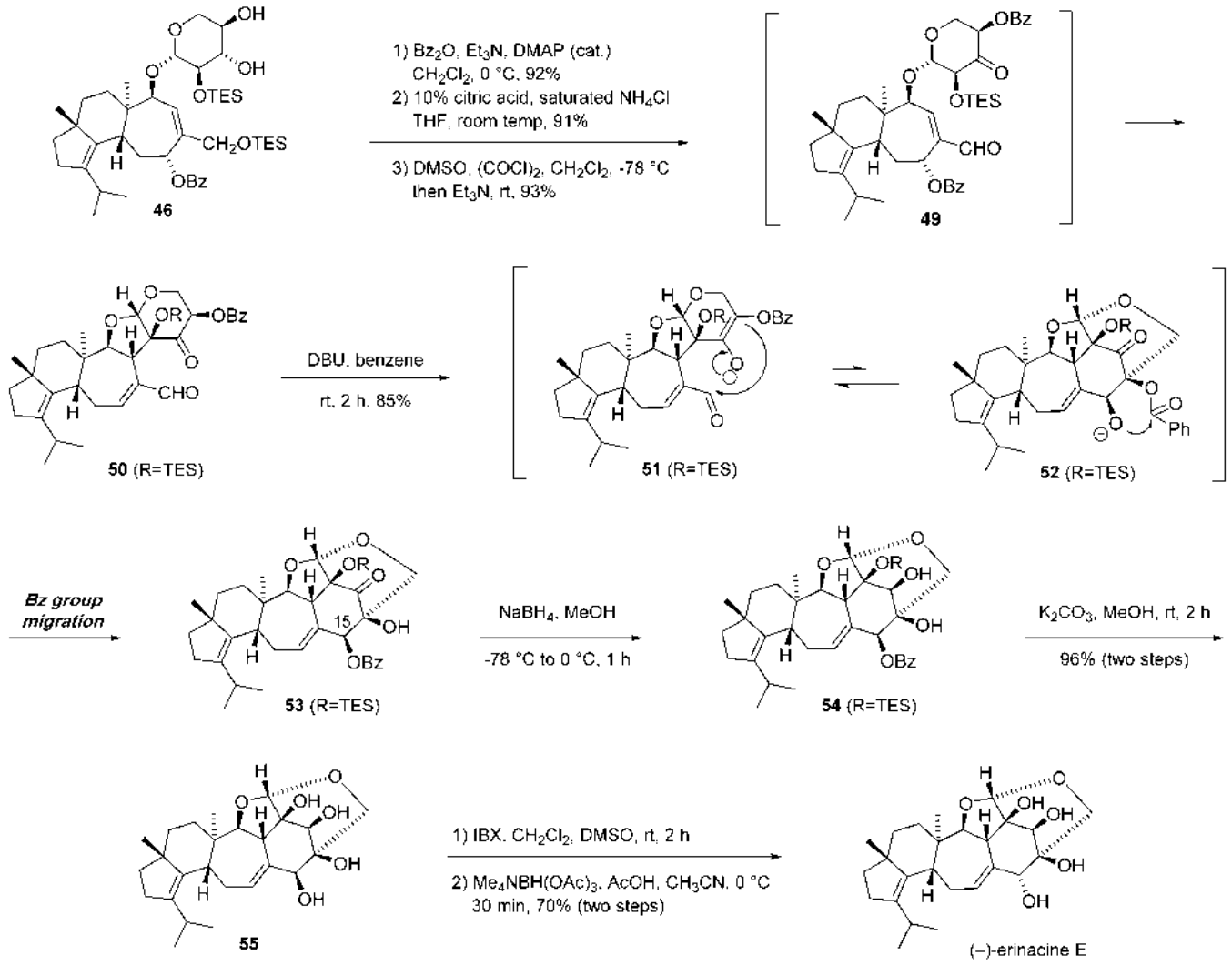

Scheme 11. Biomimetic total synthesis of (–)-erinacine E.

sequence was investigated. IBX oxidation[33] of **55** gave the desired enone; however, the reactions using a range of reducing reagents resulted in decomposition or formation of the 1,4-reduction product owing to its *s-cis* structure.

After rigorous investigations, the reduction using $Me_4NBH(OAc)_3$[34] was found to be successful and gave the desired product as a single diastereomer, which was identical in all respects to the natural product, as confirmed by the spectral data.

As described above, we have reported a highly stereoselective total synthesis of (–)-erinacine E, which was achieved in 12 steps starting from the enantiopure alcohol **35**. The intramolecular aldol reaction of ketone **50**, driven by the rationally designed benzoyl group migration, is the key step in this synthesis, which effectively prevented the retro-aldol reaction to permit the successful construction of the strained structure of (–)-erinacine E.

The structure of a putative biosynthetic intermediate, striatal A, suggests that the C4′ acetyl group driven intramolecular aldol reaction could be involved in the biogenesis of (–)-erinacine E. The acyl group migratory intramolecular aldol reaction could be applied to the synthesis of other strained molecules.

Total Synthesis of (–)-Scabronine G and A, and (–)-Episcabronine A[14g]

(–)-Scabronines A–G[6a,b,d] and (–)-episcabronine A[6c] were isolated from the bitter mushroom *Sarcodon scabrosus* and

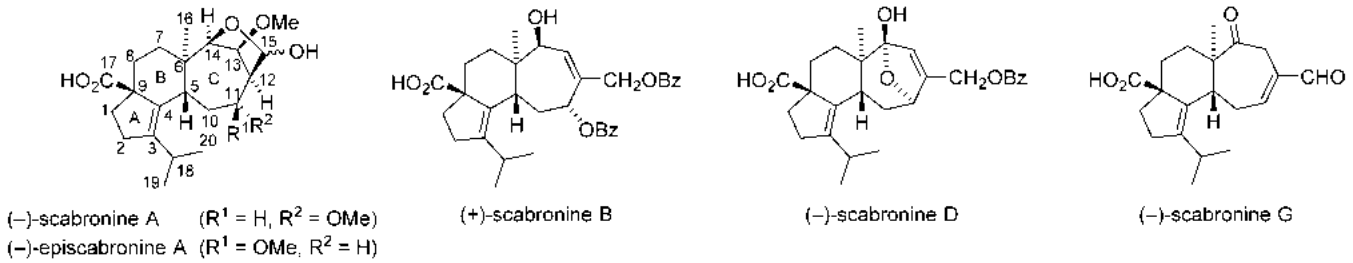

Fig. 3. Structures of (–)-scabronine A, (–)-episcabronine A, (+)-scabronine B, and (–)-scabronines D and G.

characterized by Ohta and co-workers (Figure 3). (–)-Scabronine A[6a] contains six contiguous stereogenic centers in its seven-membered ring, distinguishing it as the most complex structure in the scabronine family. (–)-Scabronine G[6b] possesses the least oxidized seven-membered ring of the known scabronines, which vary in oxidation state between (–)-scabronines A and G. Biogenetic deviation of (–)-scabronine A from G through various oxidation processes is inferred by the array of isolated natural products.

(–)-Scabronines A and G demonstrate a potent induction effect upon the production and excretion of neurotrophic factors including NGF in 1321N1 human astroglial cells.[6] Accordingly, their therapeutic relevance and synthetic challenge have evoked intense investigation towards their total synthesis by a number of research groups.

However, despite the large number of synthetic studies involving the cyathane diterpenoids reported over the last decade,[13–16] only two total syntheses of the scabronines have been achieved.[14e,f] The lack of literature precedent may be attributable to their highly oxygenated structures, which are closely related to the [5-6-7] tricyclic core of cyathane diterpenoids which contain 1,4-*anti* quaternary methyl groups.

However, the C9 methyl group and/or C3 isopropyl group is oxidized in the scabronines, which distinguishes them as difficult synthetic targets. Although scabronine A is one of the most potent NGF synthesis stimulators, its total synthesis has not been achieved to date.[35] Hence, we started the enantioselective total syntheses of scabronines G and A and episcabronine A.

Our retrosynthetic analysis of (–)-scabronines G and A and (–)-episcabronine A via the key intermediate **58** is shown in Scheme 12. Considering the same relative configurations at C11 and C14 in (–)-scabronine A and (+)-scabronine B, we surmised that (+)-scabronine B is a biosynthetic intermediate of (–)-scabronine A.

Hence, we envisioned that (–)-scabronine A could be synthesized via a biomimetic approach from compound **56**, which possesses the same relative configurations of C11 and C14 as those in (+)-scabronine B. An oxa-Michael reaction at C13 from the less sterically hindered β face of **56** would form an enolate, whose subsequent protonation/acetalization cascade will give a hemiacetal with the installation of the C12 and C13 stereogenic centers. The C11 hydroxyl group in **56** was expected to resist β-elimination because of its poor leaving ability.

However, β-elimination of the C11 benzoate in **57** would follow the oxa-Michael reaction of **57**, resulting in a second oxa-Michael reaction at the C11 position (upon treatment with a strong base such as sodium methoxide), followed by a protonation/acetalization cascade to give a hemiacetal with the C11–C13 stereotriad present in (–)-episcabronine A. Therefore, (–)-episcabronine A could potentially be synthesized from **57** via a biomimetic approach.

Compound **58** was therefore a rational intermediate for the syntheses of (–)-scabronines G and A and (–)-episcabronine A, given our total syntheses of cyathanes in which the requisite stereoselective transformations from **58** to both **56** and **57** were achieved as described previously. Compound **58** was accessible from **59** by 1,2-diol cleavage and subsequent ring-closing metathesis (RCM). Compound **59** could be prepared from **60**, which could itself be generated by the oxidative dearomatization[36] of compound **61** and subsequent intramolecular inverse electron demand Diels–Alder (IEDDA) reaction.[37]

The intramolecular IEDDA reaction was an appealing method for the simultaneous generation of two stereogenic centers (C6 and C9) and the C3–C4 tetrasubstituted alkene. Considering the transition states of the intramolecular IEDDA reaction, we set the hydroxyl and methoxy groups at suitable positions of the benzene ring in **61** to render the intramolecular IEDDA reaction stereoselective. Chiral allene **61** could be prepared from **62** with concomitant chirality transfer.

The total syntheses of (–)-scabronines G and A and (–)-episcabronine A began with the preparation of **62** (Scheme 13). TIPS protection of the known compound **63**[38] gave TIPS ether **64** in 83% yield. Reaction of **64** with a propargylzinc reagent in the presence of a catalytic amount of titanium tetrachloride[39] afforded an intermediate alcohol, whose reductive removal using triethylsilane gave alkyne **65** in 84% yield over two steps. Lithiation of the terminal alkyne and subsequent reaction with Weinreb amide **66**[40] provided ketone **67** in 89% yield.

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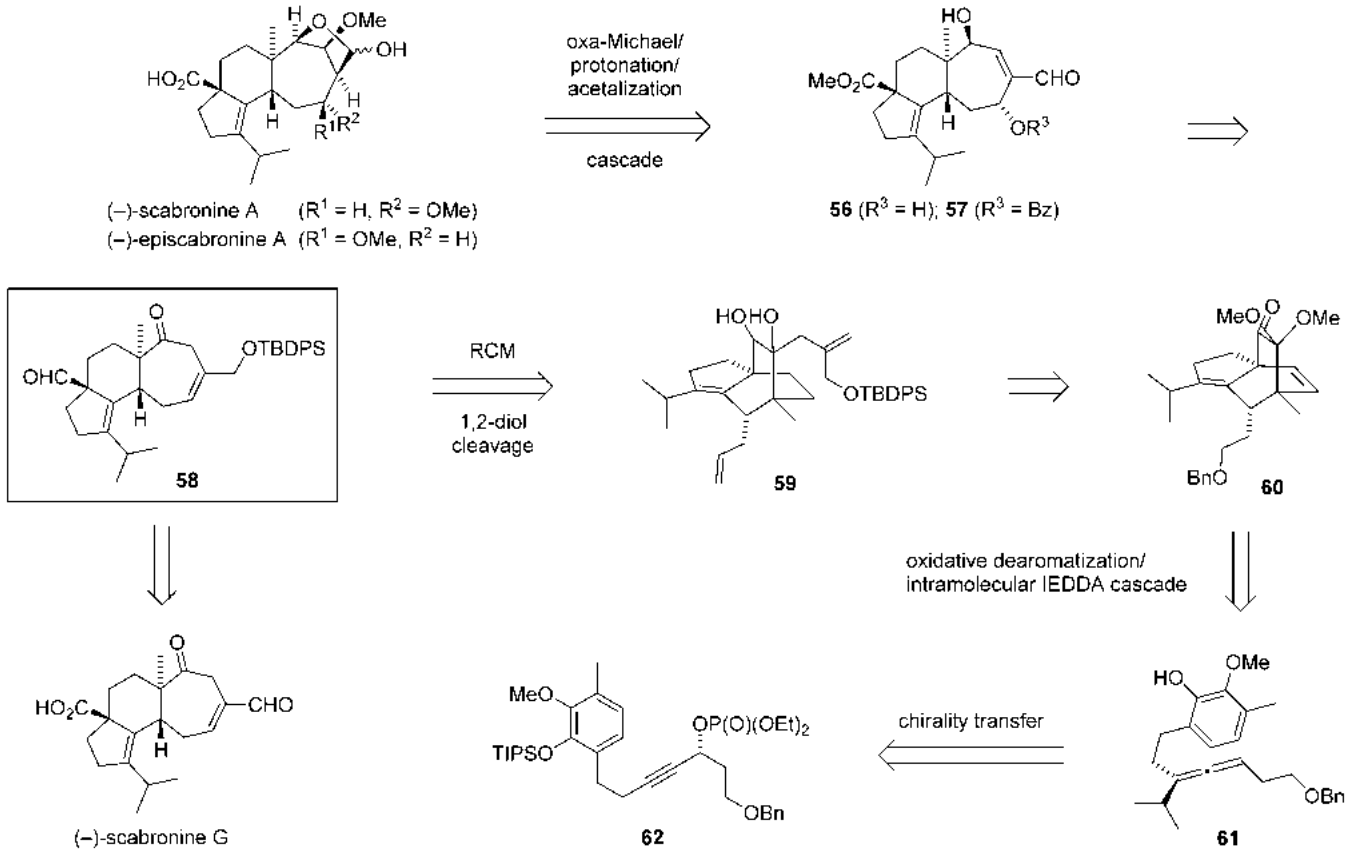


Scheme 12. Retrosynthetic analysis of (–)-scabronines G and A, and (–)-episcabronine A.

Noyori reduction[41] of **67** using 6 mol% of the Ru[(*R,R*)-Tsdpen](*p*-cymene) catalyst gave the desired alcohol in 95% *ee*, which was converted to phosphate **62** in 89% yield over two steps. The reaction of **62** with isopropylmagnesium chloride and CuCN·2LiCl[42] gave the chiral allene in quantitative yield, which was followed by TIPS deprotection to give **61** in 97% yield. HPLC analysis of **61** confirmed complete chirality transfer from **62** to chiral allene **61**.

The reaction of **61** with phenyliodine(III) diacetate (PIDA) afforded an *o*-benzoquinone mono-dimethylacetal within 15 min at 0°C. The subsequent intramolecular IEDDA proceeded slowly at room temperature and required seven days to go to completion to give **60** in 97% yield. Benzyl deprotection of **60** gave alcohol **68** in 92% yield and 95% *ee*, whose structure was confirmed by X-ray crystallographic analysis.

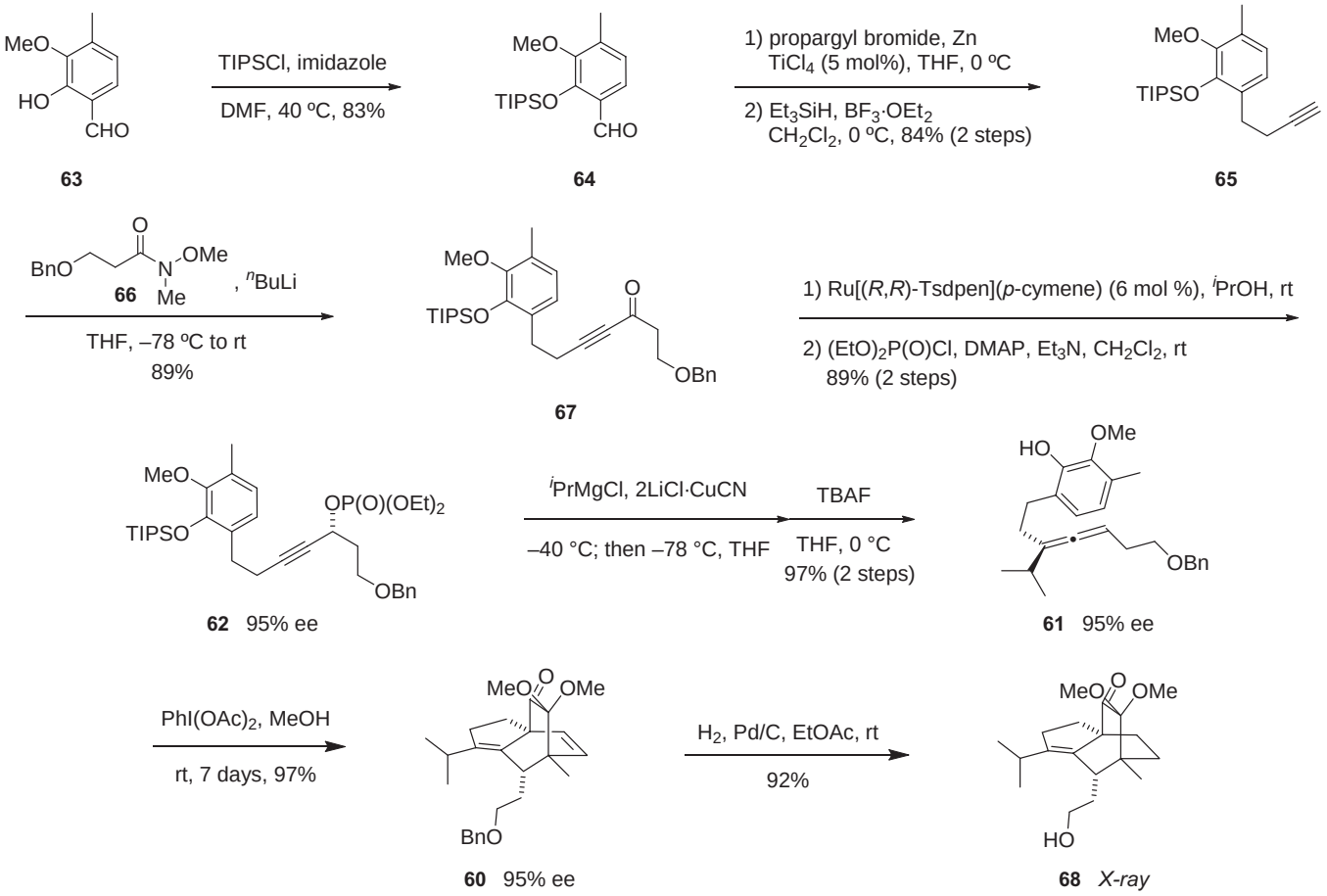
The high stereoselectivity of this reaction was attributed to the IEDDA transition state, in which the steric repulsion between the dimethyl acetal and 2-benzyloxyethyl groups is minimized (Scheme 14). The two interconvertible transition states **61a** and **61b** are possible, but it is transition state **61a** that leads to the desired product **60** and avoids steric repulsion between the acetal and benzyloxyethyl groups.

Swern oxidation of **68** and Wittig reaction of the resulting aldehyde gave **69**, which was reduced using sodium borohydride to give hydroxy ketone **70** after acidic workup in 92% yield over three steps (Scheme 15). Compound **70** was added to an organozinc reagent[43] (prepared in situ) to give *cis*-1,2-diol **59** as a single isomer in 91% yield. NOESY analysis of brosylate derivative **71** confirmed the structure shown in Scheme 15.

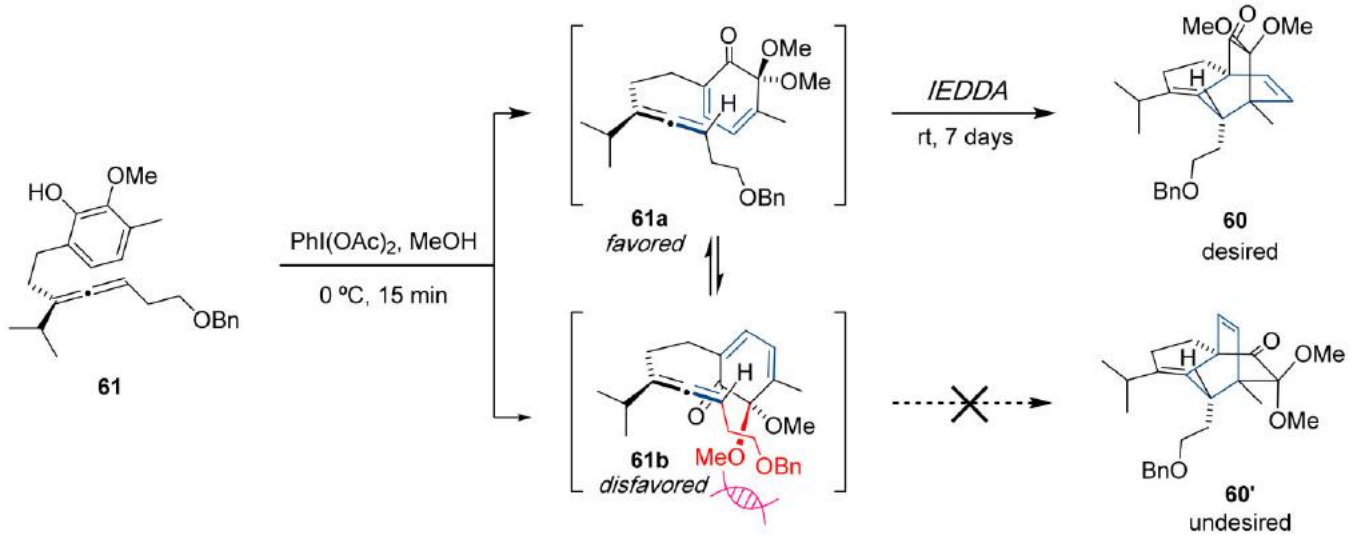
Compound **59** resisted oxidative cleavage by $NaIO_4$, probably as a result of steric hindrance. However, upon reaction with PIDA,[44] successful oxidative cleavage gave an intermediate keto aldehyde that underwent RCM using Grubbs II catalyst[45] to give the key intermediate **58** in 88% yield over two steps.

TBDPS deprotection of **58** using TBAF/AcOH with subsequent Pinnick oxidation of the aldehyde in the newly formed alcohol **73** gave hydroxy carboxylic acid **74** in 68% yield over two steps (Scheme 16). Finally, Dess–Martin oxidation of **74** gave pure scabronine G in an overall 21% yield over 19 steps (from **63**), which proved to be identical to naturally occurring (–)-scabronine G[6b] in all respects.

We next investigated the total synthesis of (–)-scabronine A and its methyl ester from compound **56** (Scheme 17). In an



Scheme 13. Highly stereoselective preparation of **68**.



Scheme 14. Proposed transition states of the IEDDA.

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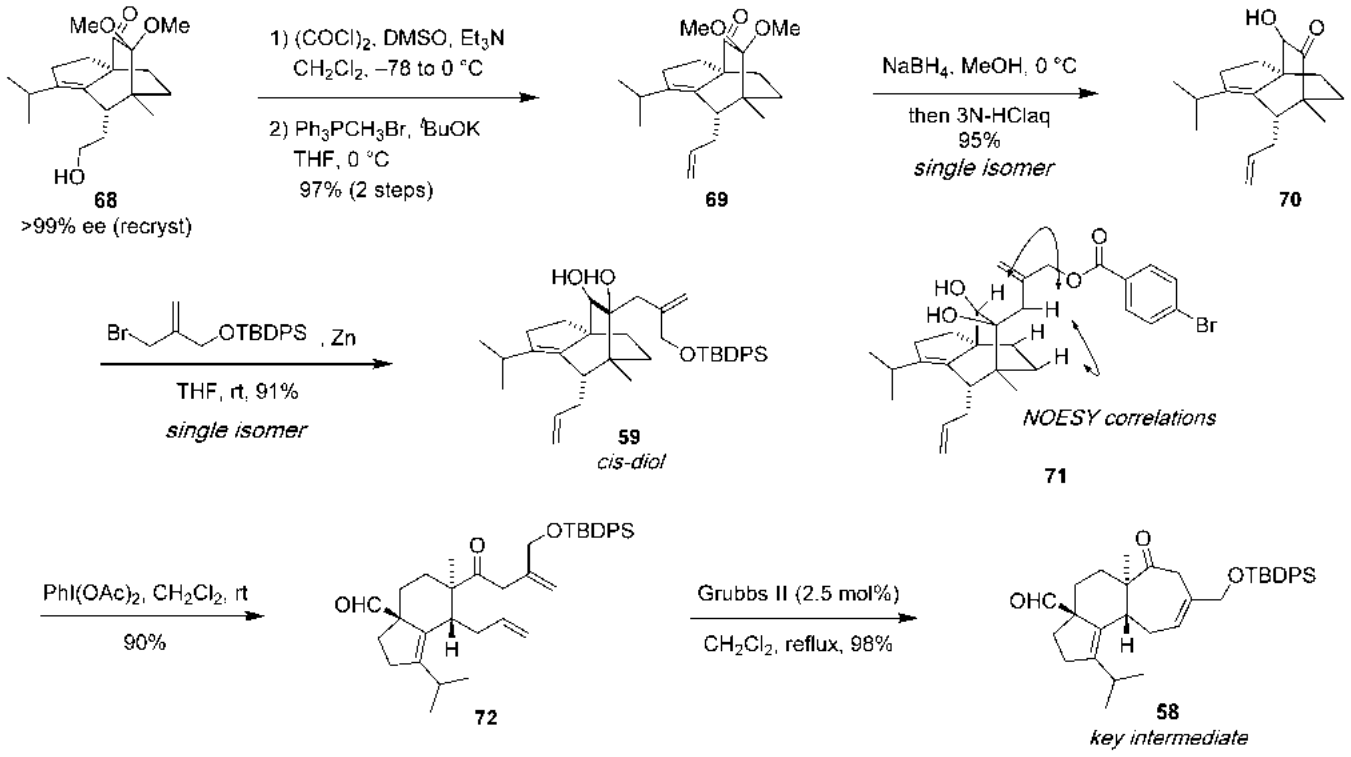
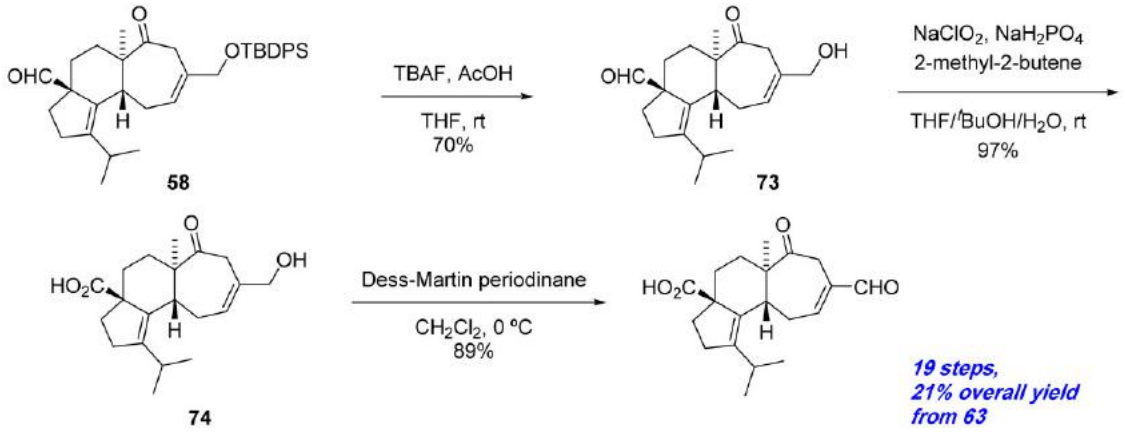


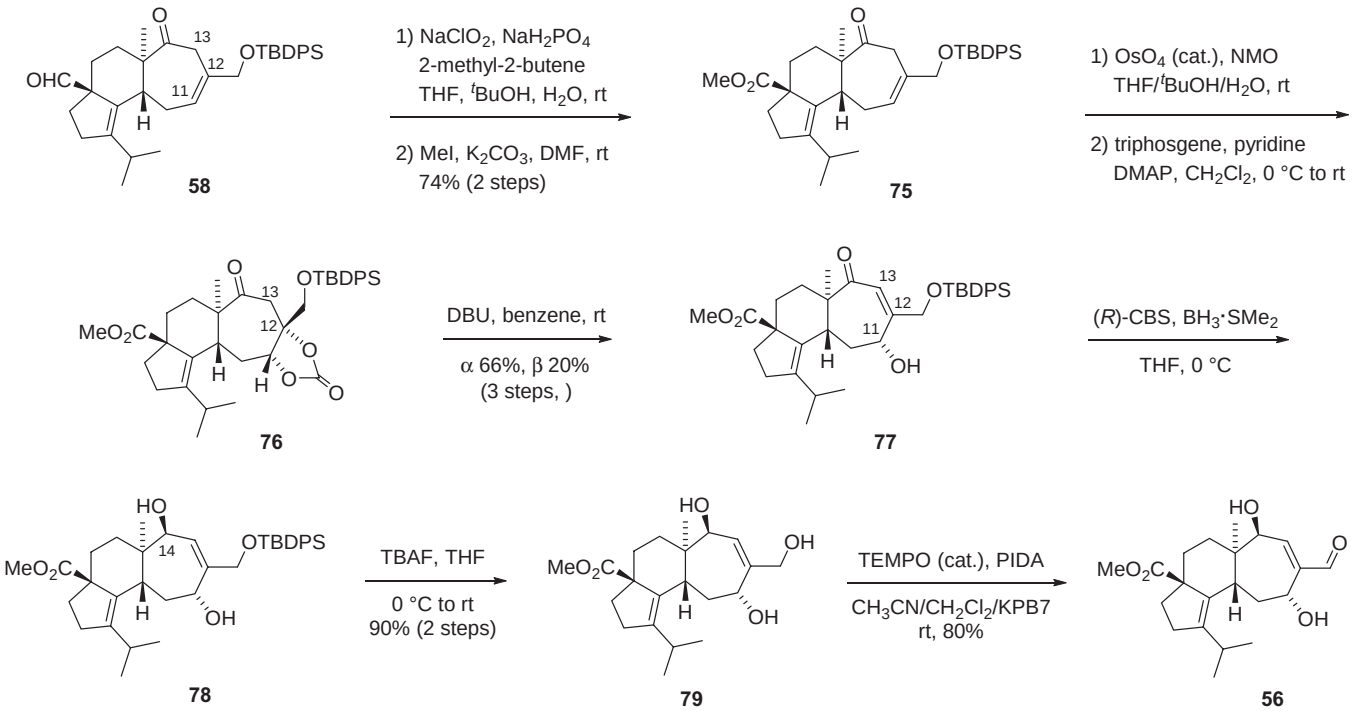
Scheme 15. Synthesis of key intermediate **58**.



Scheme 16. Enantioselective total synthesis of (–)-scabronine G.

effort to prepare **56**, direct epoxidation of the C11–C12 alkene in **58** and its derivatives was attempted but proved unsuccessful. However, the corresponding methyl ester, prepared by the Pinnick oxidation of **58** and subsequent methylation of the resulting carboxylic acid, was successfully converted to compound **76** (dr = 3.3:1) by conventional dihydroxylation and carbonate formation. Treatment of **76** with DBU led to clean β-elimination to give **77** in 49% yield over five steps from **58**. CBS reduction of ketone **77**,[14i,28] TBDPS deprotection of the resulting diol **78** using TBAF, and selective oxidation of the primary hydroxyl in compound **79** using TEMPO/PIDA gave **56** as a single isomer in 72% yield over three steps.

Armed with the necessary precursor, the stage was set for an oxa-Michael/protonation/acetalization cascade. The reaction of **56** with sodium methoxide in methanol produced compound **80**, which was treated directly with acidic methanol



Scheme 17. Enantioselective preparation of **56**.

to afford compound **81** as a mixture of C15 isomers (Scheme 18).[46] No side products were observed in this cascade reaction. The stereoselectivity of the oxa-Michael reaction and subsequent protonation of the enolate was attributed to the conformational bias and directing effect of the C14 and the C11 hydroxyl groups, respectively. The C11 hydroxyl group of **81** was methylated under conventional conditions and the product converted to either (–)-scabronine A (94%) or its methyl ester, which proved to be identical in all respects to the naturally occurring compounds according to the spectral data.[6a]

Subsequently, we examined another planned cascade reaction from **57**. For this purpose, compound **57** was prepared from **75** in four steps, as shown in Scheme 19. That is, benzoylation of **75** afforded **82**, followed by CBS reduction and selective TEMPO oxidation under optimized conditions using potassium phosphate buffer (pH = 7.0) (KPB7) to deliver compound **57**.

In an analogous cascade reaction to that shown in Scheme 18, treating **57** with sodium methoxide gave **84** as a single isomer (Scheme 20). The highly stereoselective transformations involved an oxa-Michael reaction of **57** to install the β-methoxy group at the C13 position to afford an enolate, followed by β-elimination of the C11 benzoate to generate an enal. Subsequently, the enal underwent a second oxa-Michael reaction to install the β-methoxy group at the C11 position, followed by protonation at C12 from the less sterically hindered α face. A final acetalization step furnished compound **84**. The C11 benzoate of **57** was converted to a methoxy group with complete inversion of the configuration during the cascade reaction. Hydrolysis of the methyl ester in **84** gave (–)-episcabronine A, which proved to be identical in all respects to the natural product, as per the spectral data reported.[6c]

The cascade reactions in this work suggest that (–)-scabronine A and (–)-episcabronine A could be artifacts, because these compounds were isolated from the methanol extract of *Sarcodon scabrosus*. We are currently investigating the possibility of their formation during the isolation process.

Total Synthesis of (–)-Scabronine D[14o]

Oshima and Ohta et al. isolated (–)-scabronine D (Scheme 21) as a component of various scabronines from *Sarcodon scabrosus*,[6d] a mushroom belonging to the Thelephoraceae family that has a strong bitter taste. (–)-Scabronine D contains five stereogenic centers in its cyathane scaffold, which features a [5-6-7] tricyclic ring system with a *trans*-[6-7] ring fusion and two all-carbon quaternary stereogenic centers at the ring junctions.

In addition, the unique oxygen-bridged hemiacetal structure formed between the C11 and C14 positions distinguishes it as a relatively complex structure in the scabronine family. In

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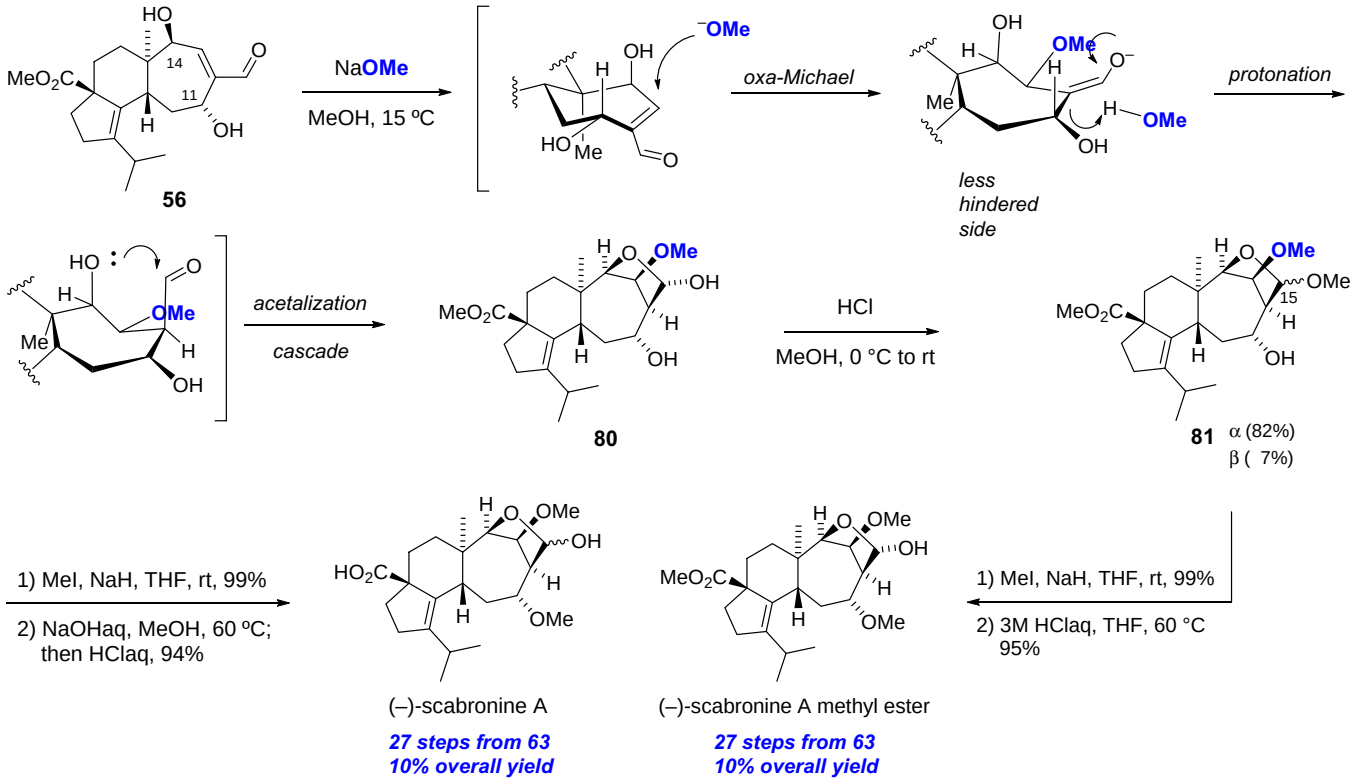
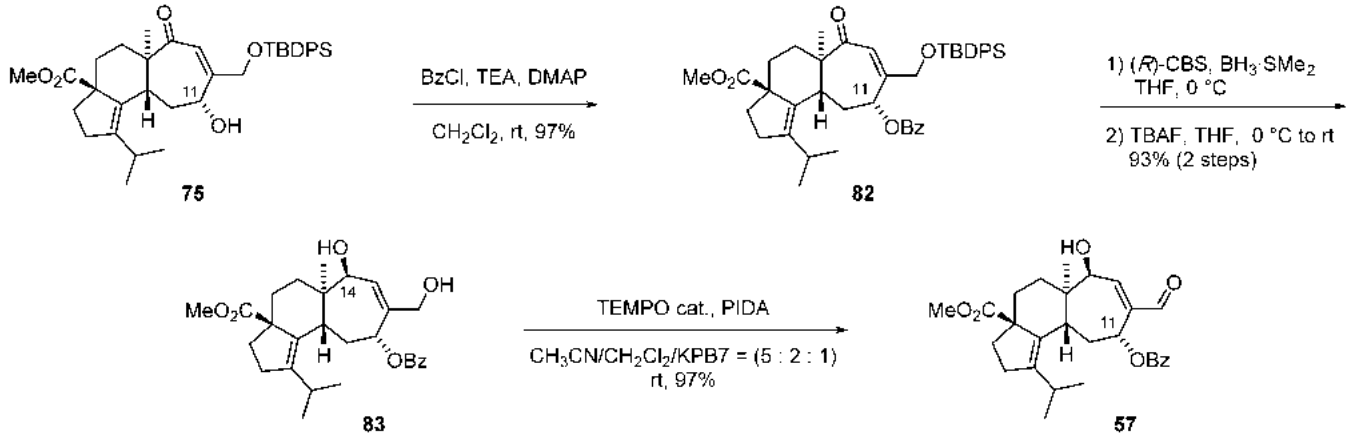


Scheme 18. Enantioselective total synthesis of (–)-scabronine A and its methyl ester via the highly stereoselective oxa-Michael/protonation/acetalization reaction cascade.

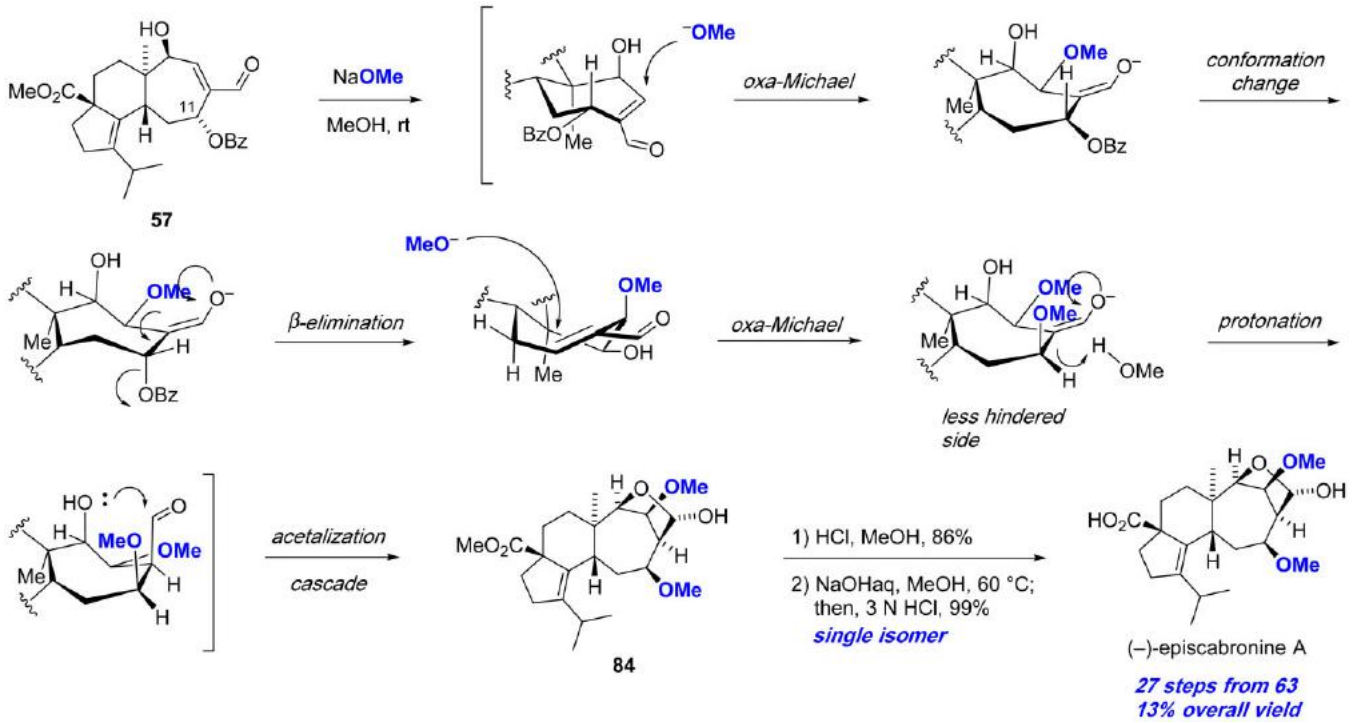


Scheme 19. Preparation of **57**.

the course of continued research on the collective and diverse total synthesis of scabronines for understanding their structure–activity relationships, we focused our interest on the structurally unique (–)-scabronine D.

TBDPS deprotection of compound **77** using TBAF gave compound **85**, which was in fast equilibrium with **85′**, with a 1:1 ratio of **85**:**85′** observed in $CDCl_3$ at 20°C (Scheme 21). To accomplish the total synthesis of (–)-scabronine D, the conversion of **85** to the natural product was examined. Hydrolysis of the methyl ester in **85** did not occur at room temperature under any of the reaction conditions investigated.

Therefore, heating was necessary for hydrolysis to occur, but using highly basic conditions resulted in the formation of undesired products derived from the decomposition of **85**. To



Scheme 20. Enantioselective total synthesis of (–)-episcabronine A via highly stereoselective oxa-Michael/β-elimination/ oxa-Michael/protonation/acetalization reaction cascade.

prevent decomposition from occurring, the reactive C14 keto and C11 hydroxyl groups in compound **85** were co-protected as the methyl acetal upon treatment with acidic methanol and gave compound **86** in 98% yield.

Transformation of compound **86** to (–)-scabronine D required converting the methyl ester into a *tert*-butyl ester because the benzoyl group is removed under the conditions required to remove the methyl ester. Therefore, methyl ester **86** was converted into carboxylic acid **87** using hydrolysis under basic conditions. Although the carboxyl group in compound **87** was sterically hindered due to the adjacent all-carbon quaternary center, the reaction of **87** with *N,N′*-diisopropyl-*O*-*tert*-butyl isourea[47] gave *tert*-butyl ester **88**.

The reaction of **88** with benzoyl chloride under conventional conditions afforded compound **89** in 90% yield over three steps. The *tert*-butyl group in **89** was effectively removed upon treatment with TFA, but the methyl acetal remained intact. Therefore, after the completion of this reaction, all volatile materials were removed in vacuo and the residue was dissolved in THF and treated with 3 N HCl at 60°C.

Fortunately, the methyl acetal could be removed stepwise via compound **90** to give (–)-scabronine D in 83% yield over two steps from **89**. Interestingly, the final product was obtained in only the hemiketal form and was identical in all respects to the spectral data reported for the natural product by Oshima and Ohta et al. However, Oshima and Ohta et al. reported (–)-scabronine D as a yellow oil, but the synthetic (–)-scabronine D prepared by the route described here solidified upon standing.

Summary and Outlook

We have achieved the first enantioselective total synthesis of allocyathin B_2, erinacine B, and erinacine E via a highly convergent approach using novel chiral building blocks prepared by the asymmetric catalysis route developed by our group. This synthetic approach incorporates the divergent synthesis of chiral building blocks by asymmetric catalysis, allowing the enantioselective total syntheses of cyathane diterpenoids and the preparation of various derivatives for structure–activity relationship studies. Further synthetic studies recently undertaken in the group feature an efficient and short access to the key intermediate used in these total syntheses, and will be reported in the near future.

The enantioselective total syntheses of scabronines G, A, D, and episcabronine A were achieved. These syntheses feature the stereoselective construction of the AB ring system of the scabronines by an oxidative aromatization/intramolecular IEDDA reaction cascade. This approach enables the total syn-

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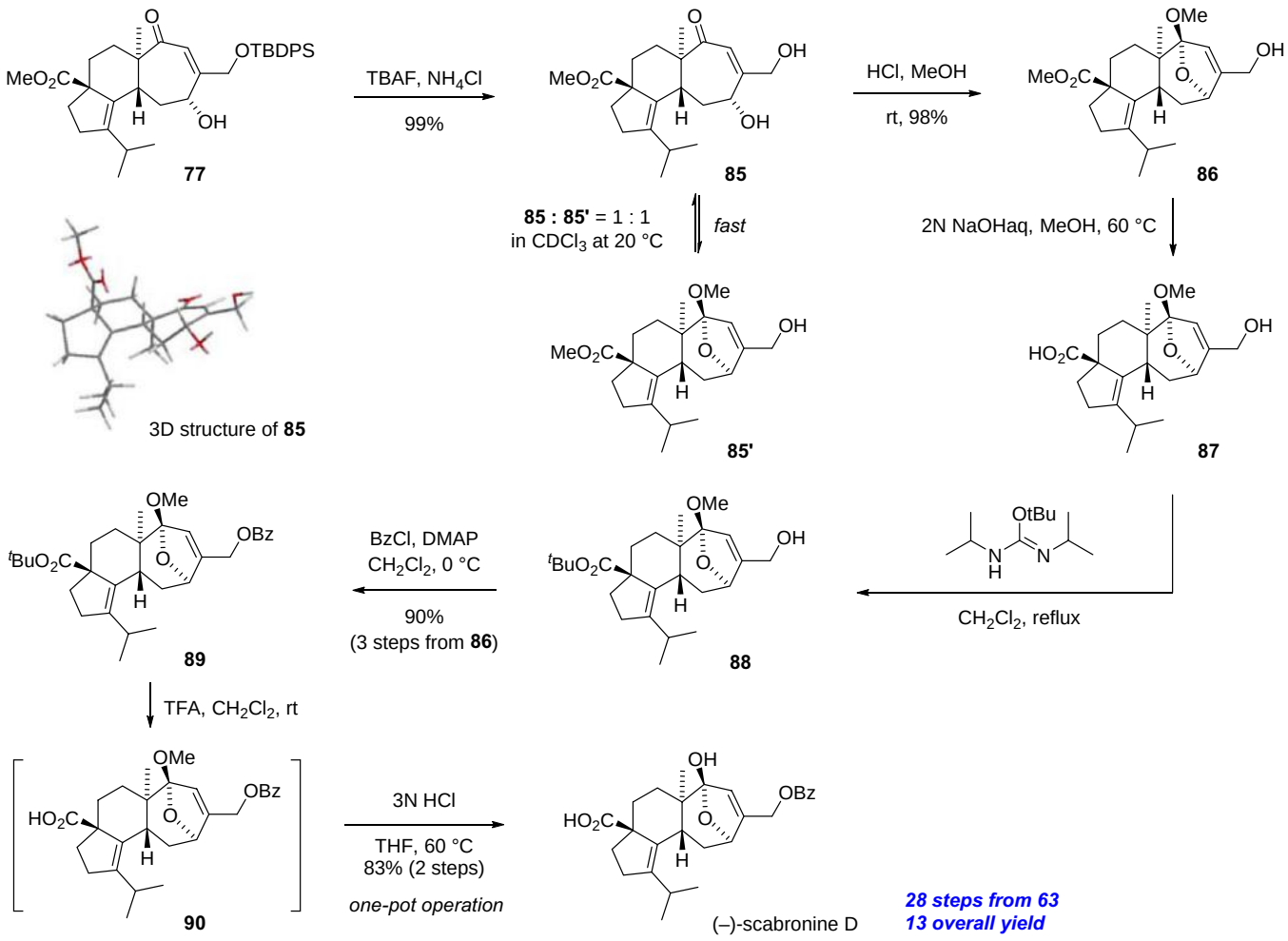

Scheme 21. Enantioselective total synthesis of (–)-scabronine D.

thesis of scabronine G in 19 steps and an overall 21% yield. The highly stereoselective oxa-Michael/protonation/acetalization cascade, inspired by the biogenetic precedent of scabronines, was realized by setting the C14 and C11 hydroxyl groups with correct configuration on the seven-membered C ring to accomplish the first total synthesis of scabronine A. Substitution of the C11 hydroxyl for the benzoate group in the substrate altered the cascade to involve a β-elimination/oxa-Michael cascade, leading to the first total synthesis of episcabronine A. The high efficiency of this divergent synthetic approach, which includes *time and space integration* steps,[48] makes it a valuable contribution to the synthesis of scabronine derivatives needed for structure–activity relationship studies.

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REFERENCES

[1] (a) B. N. Johri, H. J. Brodie, A. D. Allbutt, W. A. Ayer, H. Taube, *Experientia* **1971**, *27*, 853; (b) B. N. Johri, H. J. Brodie, *Can. J. Microbiol.* **1971**, *17*, 1243–1245; (c) A. D. Allbutt, W. A. Ayer, H. J. Brodie, B. N. Johri, H. Taube, *Can. J. Microbiol.* **1971**, *17*, 1401–1407; (d) W. A. Ayer, H. Taube, *Tetrahedron Lett.* **1972**, *19*, 1917–1920; (e) W. A. Ayer, L. L. Carstens, *Can. J. Chem.* **1973**, *51*, 3157–3160; (f) W. A. Ayer, H. Taube, *Can. J. Chem.* **1973**, *51*, 3842–3854; (g) W. A. Ayer, L. M. Browne, S. Fernandez, D. E. Ward, T. Yoshida, *Rev. Latinoam. Quim.* **1978**, *9*, 177–184; (h) H. J. Hecht, G. Hoefle, W. Steglich, T. Anke, F. Oberwinkler, *J. Chem. Soc., Chem. Commun.* **1978**, 665–666; (i) W. A. Ayer, L. M. Browne, J. R. Mercer, D. R. Taylor, D. E. Ward, *Can. J. Chem.* **1978**, *56*, 717–721; (j) W. A. Ayer, T. Yoshida, D. M. J. van Schie, *Can. J. Chem.* **1978**, *56*, 2113–2120; (k) W. A. Ayer, T. T. Nakashima, D. E. Ward, *Can. J. Chem.* **1978**, *56*, 2197–2199; (l) W. A. Ayer, S. P. Lee, *Can. J. Chem.* **1979**, *57*, 3332–3337; (m) H. Shibata, T. Tokunaga, D. Karasawa, A. Hirota, M. Nakayama, H. Nozaki, T. Tada, *Agric. Biol. Chem.* **1989**, *53*, 3373–3375; (n) B.-J. Ma, J.-K. Liu, *J. Basic Microbiol.* **2005**, *45*, 328–330; (o) Y. Shiono, F. Hiramatsu, T. Murayama, T. Koseki, T. Funakoshi, *Chem. Biodiversity* **2008**, *5*, 1811–1816; (p) S.-T. Fang, L. Zhang, Z.-H. Li, B. Li, J.-K. Liu, *Chem. Pharm. Bull.* **2010**, *58*, 1176–1179; (q) S.-T. Fang, T. Feng, L. Zhang, Z.-J. Domg, Z.-H. Li, J.-K. Liu, *Nat. Prod. Bioprospect.* **2011**, *1*, 37–40; (r) J. J. Han, Y. H. Chen, L. Bao, X. L. Yang, D. Liu, S. J. Li, F. Zhao, H. Liu, *Fitoterapia* **2013**, *84*, 22–31; (s) Z. Xu, S. Yan, K. Bi, J. Han, Y. Chen, Z. Wu, Y. Chen, H. Liu, *Fitoterapia* **2013**, *86*, 159–162; (t) B. Wang, J. Han, W. Xu, Y. Chen, H. Liu, *Food Chem.* **2014**, *152*, 169–176.

[2] H. Shibata, A. Irie, Y. Morita, *Biosci., Biotechnol., Biochem.* **1998**, *62*, 2450–2452.

[3] T. Anke, F. Oberwinkler, W. Steglich, G. Hoefle, *J. Antibiot.* **1977**, *30*, 221–225.

[4] M. Hirota, K. Morimura, H. Shibata, *Biosci., Biotechnol., Biochem.* **2002**, *66*, 179–184.

[5] (a) H. Kawagishi, A. Shimada, R. Shirai, K. Okamoto, F. Ojima, H. Sakamoto, Y. Ishiguro, S. Furukawa, *Tetrahedron Lett.* **1994**, *35*, 1569–1572; (b) H. Kawagishi, A. Shimada, K. Shizuki, H. Mori, K. Okamoto, H. Sakamoto, S. Furukawa, *Heterocycl. Commun.* **1996**, *2*, 51–54; (c) D. Mebs, T. Omori-Satoh, Y. Yamakawa, Y. Nagaoka, *Toxicon* **1996**, *34*, 1313–1316; (d) H. Kawagishi, A. Shimada, S. Hosokawa, H. Mori, H. Sakamoto, Y. Ishiguro, S. Sakemi, J. Bordner, N. Kojima, S. Furukawa, *Tetrahedron Lett.* **1996**, *37*, 7399–7402; (e) T. Saito, F. Aoki, H. Hirai, T. Inagaki, Y. Matsunaga, T. Sakakibara, S. Sakemi, Y. Suzuki, S. Watanabe, O. Suga, T. Sujaku, A. A. Smogowicz, S. J. Truesdell, J. W. Wong, A. Nagahisa, Y. Kojima, N. Kojima, *J. Antibiot.* **1998**, *51*, 983–990; (f) T. Omori-Satoh, Y. Yamakawa, D. Mebs, *Toxicon* **2000**, *38*, 1561–1580; (g) E. W. Lee, K. Shizuki, S. Hosokawa, M. Suzuki, H. Suganuma, T. Inakuma, J. Li, M. Ohnishi-Kameyama, T. Nagata, S. Furukawa, H. Kawagishi, *Biosci., Biotechnol., Biochem.* **2000**, *64*, 2402–2405; (h) H. Kenmoku, T. Sassa, N. Kato, *Tetrahedron Lett.* **2000**, *41*, 4389–4393; (i) H. Kenmoku, N. Kato, M. Shimada, M. Omoto, A. Mori, W. Mitsuhashi, T. Sassa, *Tetrahedron Lett.* **2001**, *42*, 7439–7442; (j) H. Kenmoku, T. Shimai, T. Toyomasu, N. Kato, T. Sassa, *Biosci., Biotechnol., Biochem.* **2002**, *66*, 571–575; (k) T. Kamo, Y. Imura, T. Hagio, H. Makabe, H. Shibata, M. Hirota, *Biosci., Biotechnol., Biochem.* **2004**, *68*, 1362–1365; (l) H. Kenmoku, K. Tanaka, K. Okada, N. Kato, T. Sassa, *Biosci., Biotechnol., Biochem.* **2004**, *68*, 1786–1789; (m) H. Kawagishi, A. Masui, S. Tokuyama, T. Nakamura, *Tetrahedron* **2006**, *62*, 8463–8466; (n) B.-J. Ma, Y. Zhou, L.-Z. Li, H.-M. Li, Z.-M. Gao, Y. Ruan, *Z. Naturforsch., B: J. Chem. Sci.* **2008**, *63*, 1241–1242.

[6] Scabronine A: (a) T. Ohta, T. Kita, N. Kobayashi, Y. Obara, N. Nakahata, Y. Ohizumi, Y. Takaya, Y. Oshima, *Tetrahedron Lett.* **1998**, *39*, 6229–6232. Scabronine G: (b) Y. Obara, H. Kobayashi, T. Ohta, Y. Ohizumi, N. Nakahata, *Mol. Pharmacol.* **2001**, *59*, 1287–1297. Episcabronine A: (c) T. Ohta, H. Kobayashi, S. Hosoi, F. Kiuchi, T. Kita, Y. Takaya, Y. Oshima, Y. Obara, N. Nakahata, Y. Ohizumi, *Tennen Yuki Kagobutsu Toronkai Koen Yoshishu* **1998**, *40*, 341–346. Scabronines B–F: (d) T. Kita, Y. Takaya, Y. Oshima, T. Ohta, K. Aizawa, T. Hirano, T. Inakuma, *Tetrahedron* **1998**, *54*, 11877–11886; (e) Y. Obara, N. Nakahata, T. Kita, Y. Takaya, H. Kobayashi, S. Hosoi, F. Kiuchi, T. Ohta, Y. Oshima, Y. Ohizumi, *Eur. J. Pharmacol.* **1999**, *370*, 79–84; (f) B.-J. Ma, H.-J. Zhu, J.-K. Liu, *Helv. Chim. Acta* **2004**, *87*, 2877–2881; (g) B.-J. Ma, Y. Ruan, *J. Antibiot.* **2008**, *61*, 86–88; (h) X.-W. Shi, L. Liu, J.-M. Gao, A.-L. Zhang, *Eur. J. Med. Chem.* **2011**, *46*, 3112–3117; (i) X.-W. Shi, A.-L. Zhang, G. Pescitelli, J.-M. Gao, *Chirality* **2012**, *24*, 386–390; (j) L. Liu, X.-W. Shi, S.-C. Zong, J.-J. Tang, J.-M. Gao, *Bioorg. Med. Chem. Lett.* **2012**, *22*, 2401–2406.

[7] (a) M. Curini, F. Maltese, M. C. Marcotullio, L. Menghini, R. Pagiotti, O. Rosati, G. Altinier, A. Tubaro, *Planta Med.* **2005**, *71*, 194–196; (b) M. C. Marcotullio, R. Pagiotti, V. Campagna, F. Maltese, G. Fardella, G. Altinier, A. Tubaro, *Nat. Prod. Res.* **2006**, *20*, 917–921; (c) M. C. Marcotullio, R. Pagiotti, F. Maltese, G. Norbert, O.-M. Mwankie, T. Hoshino, Y. Obara, N. Nakahata, *Bioorg. Med. Chem.* **2007**, *15*, 2878–2882.

[8] (a) M. C. Marcotullio, R. Pagiott, F. Maltese, Y. Obara, T. Hoshino, N. Nakahata, M. Curini, *Planta Med.* **2006**, *72*, 819–823; (b) Y. Obara, T. Hoshino, M. C. Marcotullio, R. Pagiotti, N. Nakahata, *Life Sci.* **2007**, *80*, 1669–1677; (c) M. C. Marcotullio, O. Rosati, F. Maltese, F. Messina, *Rec. Nat. Prod.* **2013**, *7*, 239–241.

[9] (a) D. Green, I. Goldberg, Z. Stein, M. Ilan, Y. Kashman, *Nat. Prod. Lett.* **1992**, *1*, 193–199; (b) J. Peng, K. Walsh, V. Weedman, J. D. Bergthold, J. Lynch, K. L. Lieu, I. A. Braude, M. Kelly, M. T. Hamann, *Tetrahedron* **2002**, *58*, 7809–7819; (c) J. Peng, S. G. Franzblau, F. Zhang, M. T. Hamann, *Tetrahedron Lett.* **2002**, *43*, 9699–9702; (d) J. Peng, M. A. Avery, M. T. Hamann, *Org. Lett.* **2003**, *5*, 4575–4578; (e) J. Peng, N. Kasanah, C. E. Stanley, J. Chadwick, F. R. Fronczek, M. T. Hamann, *J. Nat. Prod.* **2006**, *69*, 727–730.

 © 2014 The Chemical Society of Japan and Wiley-VCH, Weinheim

[10] (a) R. M. Lindsay, *J. Neurosci.* **1988**, *8*, 2394–2405; (b) M. V. Sofroniew, C. L. Howe, W. C. Mobley, *Annu. Rev. Neurosci.* **2001**, *24*, 1217–1281; (c) X. He, K. C. Garcia, *Science* **2004**, *304*, 870–875.

[11] F. Hefti, *Annu. Rev. Pharmacol. Toxicol.* **1997**, *37*, 239–267.

[12] (a) S. Rosenberg, *Annu. Rep. Med. Chem.* **1992**, *27*, 41–48; (b) S. Carswell, *Exp. Neurol.* **1993**, *124*, 36–42; (c) C. F. Bigge, P. A. Boxer, *Annu. Rep. Med. Chem.* **1994**, *29*, 13–22; (d) S. S. Riaz, D. R. Tomlinson, *Prog. Neurobiol.* **1996**, *49*, 125–143; (e) J. M. Conner, M. H. Tuszynski, *Ment. Retard. Dev. Disabil. Res. Rev.* **1998**, *4*, 212–222; (f) P. A. Dijkhuizen, J. Verhaagen, *Neurosci. Res. Commun.* **1999**, *24*, 1–10; (g) H. U. Saragovi, K. Burgess, *Expert Opin. Ther. Pat.* **1999**, *9*, 737–751; (h) D. Kirik, B. Georgievska, A. Bjorklund, *Nat. Neurosci.* **2004**, *7*, 105–110.

[13] For a review see: (a) D. L. Wright, C. R. Whitehead, *Org. Prep. Proced. Int.* **2000**, *32*, 307–330; (b) R. M. Wilson, S. J. Danishefsky, *Acc. Chem. Res.* **2006**, *39*, 539–549; (c) W. Krzyczkowski, E. Malinowska, F. Herold, *BioTechnologia* **2008**, 146–167; (d) J. A. Enquist Jr., B. M. Stoltz, *Nat. Prod. Rep.* **2009**, *26*, 661–680.

[14] (+)-Allocyathin B_2: (a) M. Takano, A. Umino, M. Nakada, *Org. Lett.* **2004**, *6*, 4897–4900; (b) B. M. Trost, L. Dong, G. M. Schroeder, *J. Am. Chem. Soc.* **2005**, *127*, 2844–2845; (c) B. M. Trost, L. Dong, G. M. Schroeder, *J. Am. Chem. Soc.* **2005**, *127*, 10259–10268. (+)-Cyanthiwigin U: (d) M. W. B. Pfeiffer, A. J. Phillips, *J. Am. Chem. Soc.* **2005**, *127*, 5334–5335. (–)-Scabronine G: (e) S. P. Waters, Y. Tian, Y.-M. Li, S. J. Danishefsky, *J. Am. Chem. Soc.* **2005**, *127*, 13514–13515; (f) N. Kanoh, K. Sakanishi, E. Iimori, K. Nishimura, Y. Iwabuchi, *Org. Lett.* **2011**, *13*, 2864–2867; (g) Y. Kobayakawa, M. Nakada, *Angew. Chem. Int. Ed.* **2013**, *52*, 7569–7573. (+)-Cyanthiwigin AC: (h) T. J. Reddy, G. Bordeau, L. Trimble, *Org. Lett.* **2006**, *8*, 5585–5588. (–)-Erinacine B: (i) H. Watanabe, M. Takano, A. Umino, T. Ito, H. Ishikawa, M. Nakada, *Org. Lett.* **2007**, *9*, 359–362. (–)-Cyathin A_3: (j) D. E. Ward, J. Shen, *Org. Lett.* **2007**, *9*, 2843–2846. (–)-Erinacine E: (k) H. Watanabe, M. Nakada, *J. Am. Chem. Soc.* **2008**, *130*, 1150–1151. (–)-Cyanthiwigin F: (l) J. A. Enquist Jr., B. M. Stoltz, *Nature* **2008**, *453*, 1228–1231. (–)-Cyanthiwigin B, F, G: (m) J. A. Enquist Jr., S. C. Virgil, B. M. Stoltz, *Chem. Eur. J.* **2011**, *17*, 9957–9969. (+)-Cyrneine A: (n) E. Elamparuthi, C. Fellay, M. Neuburger, K. Gademann, *Angew. Chem. Int. Ed.* **2012**, *51*, 4071–4073. (–)-Scabronine D: (o) Y. Kobayakawa, M. Nakada, *Heterocycles* **2014**, *88*, 669–677; (p) Y. Kobayakawa, M. Nakada, *J. Antibiot.* **2014**, *67*, in press, DOI: 10.1038/ja.2014.23.

[15] (+)-Erinacine A: (a) B. B. Snider, N. H. Vo, S. V. O'Neil, B. M. Foxman, *J. Am. Chem. Soc.* **1996**, *118*, 7644–7645; (b) B. B. Snider, N. H. Vo, S. V. O'Neil, *J. Org. Chem.* **1998**, *63*, 4732–4740. Allocyathin B_2: (c) M. Tori, N. Toyoda, M. Sono, *J. Org. Chem.* **1998**, *63*, 306–313. Sarcodonin G: (d) E. Piers, M. Gilbert, K. L. Cook, *Org. Lett.* **2000**, *2*, 1407–1410. Allocyathin B_3: (e) D. E. Ward, Y. Gai, Q. Qiao, *Org. Lett.* **2000**, *2*, 2125–2127; (f) D. E. Ward, Y. Gai, Q. Qiao, J. Shen, *Can. J. Chem.* **2004**, *82*, 254–267. Cyanthiwigin W, Z: (g) M. W. B. Pfeiffer, A. J. Phillips, *Tetrahedron Lett.* **2008**, *49*, 6860–6861. Cyathin A_3 and B_2: (h) K. Kim, J. K. Cha, *Angew. Chem. Int. Ed.* **2009**, *48*, 5334–5336. Cyanthiwigin A, C, G, and H: (i) C. Wang, D. Wang, S. Gao, *Org. Lett.* **2013**, *15*, 4402–4405.

[16] (a) W. A. Ayer, D. E. Ward, L. M. Browne, L. T. J. Delbaere, Y. Hoyano, *Can. J. Chem.* **1981**, *59*, 2665–2672; (b) D. E. Ward, *Can. J. Chem.* **1987**, *65*, 2380–2384; (c) K. R. Dahnke, L. A. Paquette, *J. Org. Chem.* **1994**, *59*, 885–899; (d) E. Piers, K. L. Cook, *Chem. Commun.* **1996**, *16*, 1879–1880; (e) P. Magnus, L. Shen, *Tetrahedron* **1999**, *55*, 3553–3560; (f) D. L. Wright, C. R. Whitehead, E. H. Sessions, I. Ghiviriga, D. A. Frey, *Org. Lett.* **1999**, *1*, 1535–1538; (g) K. Takeda, D. Nakane, M. Takeda, *Org. Lett.* **2000**, *2*, 1903–1905; (h) P. A. Wender, F. C. Bi, M. A. Brodney, F. Gosselin, *Org. Lett.* **2001**, *3*, 2105–2108; (i) C. Thominiaux, A. Chiaroni, D. Desmaele, *Tetrahedron Lett.* **2002**, *43*, 4107–4110; (j) M. Iwamoto, M. Miyano, M. Utsugi, H. Kawada, M. Nakada, *Tetrahedron Lett.* **2004**, *45*, 8653–8657; (k) A. Sato, T. Masuda, H. Arimoto, D. Uemura, *Org. Biomol. Chem.* **2005**, *3*, 2231–2233; (l) E. Drege, G. Morgant, D. Desmaele, *Tetrahedron Lett.* **2005**, *46*, 7263–7266; (m) K. Nakashima, N. Fujisaki, K. Inoue, A. Minami, C. Nagaya, M. Sono, M. Tori, *Bull. Chem. Soc. Jpn.* **2006**, *79*, 1955–1962; (n) J. B. Sperry, D. L. Wright, *Tetrahedron* **2006**, *62*, 6551–6557; (o) E. Drege, C. Tominiaux, G. Morgant, D. Desmaele, *Eur. J. Org. Chem.* **2006**, 4825–4840; (p) S. Kobayashi, A. Ishii, M. Toyota, *Synlett* **2008**, 1086–1090; (q) H. Watanabe, M. Nakada, *Tetrahedron Lett.* **2008**, *49*, 1518–1522; (r) L. C. Miller, J. M. Ndungu, *R. Sarpong Angew. Chem. Int. Ed.* **2009**, *48*, 2398–2402.

[17] (a) M. Honma, T. Sawada, Y. Fujisawa, M. Utsugi, H. Watanabe, A. Umino, T. Matsumura, T. Hagihara, M. Takano, M. Nakada, *J. Am. Chem. Soc.* **2003**, *125*, 2860–2861; (b) M. Honma, M. Nakada, *Tetrahedron Lett.* **2003**, *44*, 9007–9011; (c) T. Sawada, M. Nakada, *Adv. Synth. Catal.* **2005**, *347*, 1527–1532; (d) H. Miyamoto, M. Iwamoto, M. Nakada, *Heterocycles* **2005**, *66*, 61–68; (e) H. Takeda, M. Nakada, *Tetrahedron: Asymmetry* **2006**, *17*, 2896–2906; (f) H. Takeda, M. Honma, R. Ida, T. Sawada, M. Nakada, *Synlett* **2007**, 579–582; (g) R. Ida, M Nakada, *Tetrahedron Lett.* **2007**, *48*, 4855–4859; (h) M. Honma, H. Takeda, M. Takano, M. Nakada, *Synlett* **2009**, 1695–1712; (i) M. Abe, A. Saito, M. Nakada, *Tetrahedron Lett.* **2010**, *51*, 1298–1302; (j) S. Hirai, M. Nakada, *Tetrahedron Lett.* **2010**, *51*, 5076–5079; (k) S. Hirai, M. Nakada, *Tetrahedron* **2011**, *67*, 518–530; (l) T. Sawada, M. Nakada, *Tetrahedron: Asymmetry* **2012**, *23*, 350–356; (m) T. Sawada, M. Nakada, *Org. Lett.* **2013**, *15*, 1004–1007.

[18] (a) M. Iwamoto, H. Kawada, T. Tanaka, M. Nakada, *Tetrahedron Lett.* **2003**, *44*, 7239–7243; (b) H. Watanabe, M. Iwamoto, M. Nakada, *J. Org. Chem.* **2005**, *70*, 4652–4658; (c) N. Noguchi, M. Nakada, *Org. Lett.* **2006**, *8*, 2039–2042; (d) K. Asakawa, N. Noguchi, M. Nakada, *Heterocycles* **2008**, *76*, 183–190; (e) K. Asakawa, N. Noguchi, S. Takashima, M. Nakada, *Tetrahedron: Asymmetry* **2008**, *19*, 2304–2309; (f) N. Noguchi, K. Tsuna, M. Nakada, *Tetrahedron: Asymmetry* **2013**,

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24, 357–361; (g) Y. Namiki, T. Fujii, M. Nakada, *Tetrahedron: Asymmetry* **2014**, *25*, 718–724.

[19] P. K. Freeman, L. L. Hutchinson, *J. Org. Chem.* **1980**, *45*, 1924–1930.

[20] D. H. R. Barton, S. W. McCombie, *J. Chem. Soc., Perkin Trans. 1* **1975**, 1574–1585.

[21] (a) E. Hasegawa, T. Kitazume, K. Suzuki, E. Tosaka, *Tetrahedron Lett.* **1998**, *39*, 4059–4062. For extensive studies, see: (b) S. H. Chung, M. S. Cho, J. Y. Choi, D. W. Kwon, Y. H. Kim, *Synlett* **2001**, 1266–1268.

[22] L. N. Mander, S. P. Sethi, *Tetrahedron Lett.* **1983**, *24*, 5425–5428.

[23] J. H. Clark, J. M. Miller, *J. Chem. Soc., Perkin Trans. 1* **1977**, 1743–1745.

[24] D. H. R. Barton, D. J. Lester, S. V. Ley, *J. Chem. Soc., Chem. Commun.* **1978**, 130–131.

[25] (a) K. C. Nicolaou, Y.-L. Zhong, P. S. Baran, *J. Am. Chem. Soc.* **2000**, *122*, 7596–7597; (b) K. C. Nicolaou, T. Montagnon, P. S. Baran, *Angew. Chem. Int. Ed.* **2002**, *41*, 993–996; (c) K. C. Nicolaou, D. L. F. Gray, T. Montagnon, S. T. Harrison, *Angew. Chem. Int. Ed.* **2002**, *41*, 996–1000.

[26] H. Kenmoku, T. Sassa, N. Kato, *Tetrahedron Lett.* **2000**, *41*, 4389–4393.

[27] K. B. Sharpless, R. C. Michaelson, *J. Am. Chem. Soc.* **1973**, *95*, 6136–6137.

[28] E. J. Corey, R. K. Bakshi, S. Shibata, *J. Am. Chem. Soc.* **1987**, *109*, 5551–5553.

[29] H. Lönn, *J. Carbohydr. Chem.* **1987**, *6*, 301–306.

[30] L. Yan, D. Kahne, *Synlett* **1995**, 523–524.

[31] IC_{50} values of (−)-erinacine E in the binding experiments of [^{3}H]CI-977, [^{3}H]DAGO, and [^{3}H]DPDPE (1 nM) to guinea pig brain membrane were 0.8 μM, >200 μM, and >200 μM, respectively. For details, see reference [5e].

[32] T. Anke, U. Rabe, P. Schu, T. Eizenhofer, M. Schrage, W. Steglich, *Z. Naturforsch., C: J. Biosci.* **2002**, *57*, 263–271.

[33] M. Frigerio, M. Santagostino, *Tetrahedron Lett.* **1994**, *35*, 8019–8022.

[34] D. A. Evans, K. T. Chapman, E. M. Carreira, *J. Am. Chem. Soc.* **1988**, *110*, 3560–3578.

[35] For our efforts on the total synthesis of scabronine A, see reference [16q].

[36] Reviews of masked *o*-benzoquinones (MOBs) in organic synthesis: (a) C.-C. Liao, R. K. Peddinti, *Acc. Chem. Res.* **2002**, *35*, 856–866; (b) D. Magdziak, S. J. Meek, T. R. R. Pettus, *Chem. Rev.* **2004**, *104*, 1383–1430; (c) C.-C. Liao, *Pure Appl. Chem.* **2005**, *77*, 1221–1234; (d) S. P. Roche, J. A. Porco Jr., *Angew. Chem. Int. Ed.* **2011**, *50*, 4068–4093.

[37] Precedents of the intramolecular IEDDA reaction cascade: (a) S. P. Cook, C. Gaul, S. J. Danishefsky, *Tetrahedron Lett.* **2005**, *46*, 843–847; (b) S. P. Cook, A. Polara, S. J. Danishefsky, *J. Am. Chem. Soc.* **2006**, *128*, 16440–16441; (c) A. Polara, S. P. Cook, S. J. Danishefsky, *Tetrahedron Lett.* **2008**, *49*, 5906–5908; (d) G. Y. C. Leung, H. Li, Q.-Y. Toh, A. M.-Y. Ng, R. J. Sum, J. E. Bandow, D. Y.-K. Chen, *Eur. J. Org. Chem.* **2011**, 183–196.

[38] R. Tello-Aburto, A. M. Harned, *Org. Lett.* **2009**, *11*, 3998–4000.

[39] T. S. Kaufman, *J. Chem. Soc., Perkin Trans. 1* **1996**, *20*, 2497–2505.

[40] K. C. Nicolaou, K. P. Cole, M. O. Frederick, R. J. Aversa, R. M. Denton, *Angew. Chem. Int. Ed.* **2007**, *46*, 8875–8879.

[41] (a) S. Hashiguchi, A. Fujii, J. Takehara, T. Ikariya, R. Noyori, *J. Am. Chem. Soc.* **1995**, *117*, 7526–7534; (b) K.-J. Haack, S. Hashiguchi, A. Fujii, T. Ikariya, R. Noyori, *Angew. Chem. Int. Ed.* **1997**, *36*, 285–288.

[42] (a) P. Knochel, M. C. P. Yeh, S. C. Berk, J. Talbert, *J. Org. Chem.* **1988**, *53*, 2390–2392; (b) P. Knochel, W. Dohle, N. Gommermann, F. F. Kneisel, F. Kopp, T. Korn, I. Sapountzis, V. A. Vu, *Angew. Chem. Int. Ed.* **2003**, *42*, 4302–4320.

[43] R. W. Heidebrecht Jr., B. Gulledge, S. F. Martin, *Org. Lett.* **2010**, *12*, 2492–2495.

[44] K. C. Nicolaou, V. A. Adsool, C. R. H. Hale, *Org. Lett.* **2010**, *12*, 1552–1555.

[45] M. Scholl, S. Ding, C. W. Lee, R. H. Grubbs, *Org. Lett.* **1999**, *1*, 953–956.

[46] For the intermolecular oxa-Michael reaction in natural product synthesis, see: (a) M. J. Ford, J. G. Knight, S. V. Ley, S. Vile, *Synlett* **1990**, 331–332; (b) S. V. Ley, A. Armstrong, D. Díez-Martín, M. J. Ford, P. Grice, J. G. Knight, H. C. Kolb, A. Madin, C. A. Marby, S. Mukherjee, A. N. Shaw, A. M. Z. Slawin, S. Vile, A. D. White, D. J. Williams, M. Woods, *J. Chem. Soc., Perkin Trans. 1* **1991**, 667–692.

[47] (a) E. Vowinkel, *Chem. Ber.* **1967**, *100*, 16–22; (b) K. R. West, K. D. Bake, S. Otto, *Org. Lett.* **2005**, *7*, 2615–2618.

[48] S. Suga, D. Yamada, J. Yoshida, *Chem. Lett.* **2010**, *39*, 404–406.

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